DESIGN OF A 3kW HORIZONTAL AXIS WIND TURBINE FOR PROSPECTIVE LOCATIONS OF BANGLADESH

By

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CERTIFICATE OF SUBMISSION

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It is hereby declared that this thesis or any part of it has not been submitted elsewhere for the award of any degree or diploma.

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This work is dedicated to our beloved parents

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ABSTRACT

Wind energy is one of the emerging renewable energy sources in Bangladesh. In a prospective site, a significant amount of energy can be harnessed from wind. However, it is very critical to conduct a rigorous analysis of available wind resources before undertaking any wind power project. Over the years, many wind resource assessment projects have been undertaken in different locations in Bangladesh and researchers have found good potentiality of wind for energy production. Keeping this in mind, in this research a wind resource assessment has been made for nine prospective locations of Bangladesh with the help of Weibull distribution. Two important Weibull parameters such as Weibull shape factor “k” and Weibull scale factor “c” have been calculated for different methods numerically using a highly functional programming language called Python. Besides a 3kW horizontal axis wind turbine blade having a rotor diameter of 3meter is also designed using an open-source, cross-platform simulation software called QBlade. Several blade designs using different types of NACA series airfoils and NREL(National Renewable Energy Laboratory) suggested airfoils have been made to get the best performing wind turbine blade. Then blade performance has been analyzed comparatively. From the analysis, the best-performing blade design has been chosen for the design of a 3kW horizontal axis wind turbine. Later a simulation has been conducted using QBlade software for determining annual energy yield for nine prospective locations in Bangladesh. Results are showing that Inani and Parkay is the most prospective area for wind energy production compared to other locations. It is also found that blade design with NACA 4818 shows better performance in terms of starting torque & airfoil thickness and NACA 4415 & NACA 4615 in terms of Cp value. So, it can be hoped that this research will help the project developers to find the prospective location for planting wind power plants in Bangladesh.

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Abbreviations

ETF	Energy Trend Factor.
ETM	Energy Trend Method.
MOM	Method of Moment or Moment Method.
PDM	Power Density Method.
EMOJ	Emperical Method of Justus.
EMOL	Emperical Method of Lysen.
MLM	Maximum Likelihood Method.
AMLM	Alternative Maximum Likelihood Method.
SDM	Standard Daviation Method.
MEPFM	Modified Energy Pattern Factor Method.
HAWT	Horizontal Axis Wind Turbine.
VAWT	Vertical Axis Wind Turbine.
BEMT	Blade Element Moment Theory.
TSR	Tip Speed Ratio.
AOA	Angle of Attack.
PDF	Probability Density Function.
NREL	National Renewable Energy Laboratory.
NACA	National Advisory Committee for Aeronautics.
GWEC	Global Wind Energy Council.
BPDB	Bangladesh Power Development Board.

Nomenclature

$F(v)$	Cumulative Distribution Function.
$f(v)$	Probability Density Function.
v, $\bar{v}$	Mean Wind Speed.
k	Weibull Shape Factor.
c	Weibull Scale Factor.
E_{pf}	Energy Trend Factor.
σ, σ_v	Standard Deviation.
Γ	Gamma Function.
v_m	Average Wind Speed.
v_i	Wind Speed In Time Step i.
n	Wind Speed Data Points Number.
f_i	Frequency of Occurrence of Each Speed Class.
P	Wind Power.
ρ	Density of Air.
A_1	Wind Area Infront of The Turbine.
A_2	Wind Area After Passing The Wind Turbine.
v_1	Incoming Wind Velocity.
v_2	Wind Velocity After Passing Through Wind Turbine.
$\dot{m}$	Mass Flow Rate.
$\bar{F}$	Force Exerted By Air On The Disc.
v'	Velocity of Air In The Disc Plane.
C_p	Power Coefficient.
P_o	Available Power.
ω	Angular Velocity of The Blade
r	Length of The Blade.
v_∞	Free Stream Velocity.

Chapter 1

INTRODUCTION

1.1 General

We are living in a time where energy is an indispensable thing for modern civilization. Energy is the foremost medium of all the activities like photosynthesis, cyclone, earthquake, tidal waves, etc happening around the world and it also plays an important role in our daily life. More implementation of energy indicates the structural development of a country. Since modern civilization is upgrading day by day the demand of energy usage is increasing day by day.

Energy can be categorized into three main parts- Renewable Energy, Nuclear Energy, Fossil Fuel Energy. From these three energy categories, we can extract various energy sources such as oil, hydro, coal, nuclear, natural gas, etc. From these sources, we get energy products like heat, electricity, light, fuel, shaft power, etc. All of these energy sources have a positive impact on global economics and technologies as well as negative impacts. The major problem of these energy sources is their negative impact on the environment. Experts believe that energy sources like coal, nuclear, natural gas, etc can cause air pollution and the greenhouse effect.

Among the main 3 categories of energy, experts are relying on Renewable energy and proposing as the fundamental source of energy soon and as the solution of all the current energy-related problems. Renewable energy is environment-friendly and it has zero impact on the environment. With the rising demand for energy and many other reasons, the prices of energy sources like fossil fuels, nuclear fuels are increasing day by day. So, people are trying to find alternative sources of energy to exploit them at the cheapest cost. Wind energy is a kind of Renewable energy source, which will never be diminished. It is available almost every time, at all places and in large quantities all over the world.

1.2 Wind as an Energy Source

The world is now highly dependable on Fossil fuels, which are the non-renewable source. Fossil fuels are finite that will eventually dwindle in the future. They are expensive and not good for the environment. Non-renewable energy sources have a greater impact on global warming. That's why developed countries, as well as developing countries, are now focusing on Renewable Energy sources such as wind energy because of its clean energy, it's free and it has a lower impact on the environment. According to the Global Wind Energy Council (GWEC) report, about 539,581 MW capacity of wind power has been installed globally at the end of 2017. It was 487,657MW at the end of 2016. Renewable Energy such as wind energy will not run out because it is replenishing constantly and it can be used generation to generation. It is one of the emerging sources of Renewable Energy. For a developing country like Bangladesh, it can hold a good prospect if the wind resource can be exploited in proper planning.

1.3 Problem Statement

We are living in a time where energy is an indispensable element for modern civilization. More implementation of energy indicates the structural development of a country. Bangladesh developing power plants to produce electricity to meet national needs. But most of these power plants are coal, oil and gas-based power plants which are pretty expensive to run. As a result, it is hard to bear all the expense for a developing country like Bangladesh These power plants have environmental issues too. Regarding all these issues, implementing Renewable source-based power plants in Bangladesh such as hydro, wind, solar, etc can reduce the great content of these problems. These Renewable sources will not extinct because they are replenishing continuously. In this study, our focus was on "Wind Energy" which can be a prospective source for energy production for Bangladesh in the future.

1.4 Objectives

The objectives of this thesis are:

1. to determine Weibull shape and scale parameters for selected locations in Bangladesh using diversified methods.
2. to design a smaller-capacity horizontal axis wind turbine for the prospective locations of Bangladesh.
3. to determine the annual energy yield from designed turbine for selected prospective locations in Bangladesh.

1.5 Scope of the Thesis

In **Chapter 1,** a brief discussion about wind energy and its advantages have been discussed. Besides our aim and research objectives has been represented.
In **Chapter 2,** global wind energy installation and wind energy installation in Bangladesh have been discussed briefly. The theory of Weibull distribution and also the method to determine the shape and scale parameter of Weibull distribution have been described in detail. Besides wind turbine types and theory related to wind turbine design have been described in detail. There are some design parameter of wind turbine that has been described briefly.
In **Chapter 3,** the methodology of our research have been discussed. Different types of airfoil and blade design geometry have been illustrated.
In **Chapter 4,** results and findings of our thesis have been discussed.
In **Chapter 5,** conclusion and future recommendation statements are given on behalf of our research.

Chapter 2

LITERATURE REVIEW

2.1 Global Scenario of Wind Energy

Many developed countries and developing countries are now producing electricity from wind energy to meet national demand. The production and installation of wind energy increasing day by day globally. According to the Global Wind Energy Council (GWEC), in the last 5 years, global installation of wind power capacity has been increased to 256,994MW. Recent statistical information about global wind energy installation is shown below-

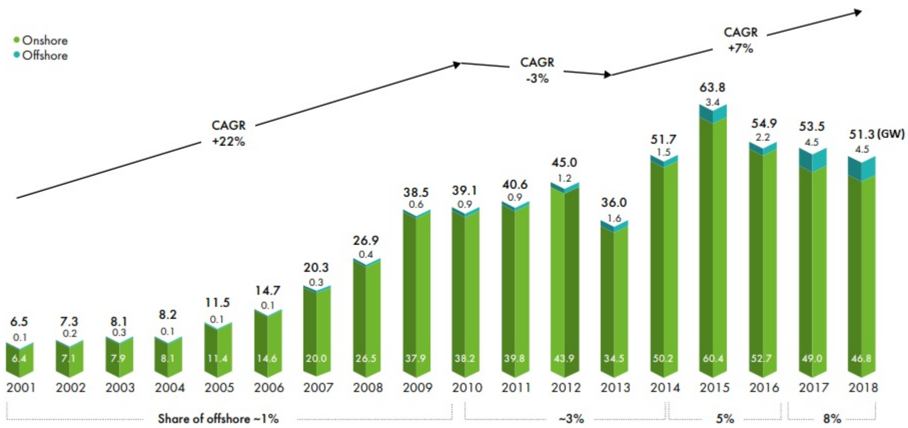

Figure 2.1: GWEC Global Wind Report 2018 (https://gwec.net/)

2.2 Wind Power Installation Scenario of Bangladesh

Bangladesh has installed several wind power plants in recent years. Bangladesh has installed both off-grid and on-grid type wind power plants that are producing energy regularly. BPDB has installed two 1MW capacity off-grid hybrid wind power plant both in Kutubdia Upazila, Cox's Bazar. Both power plants are still running. BPDB also installed 0.90MW on-grid wind power plant in Sonagazi, Feni which is still running. Besides, some on-grid wind power plant projects are under planning. They are 10MW wind power plant in Kalapara Upazila, Patuakhali by RPCL, 2MW wind power plant in Sirajganj by BPDB and 60MW wind power project at Chakaria Upazila, Cox's Bazar by US-DK green energy (BD) Ltd and BPDB.

2.3 Weibull Distribution

Azad and Alam (2012) mentioned that the Weibull distribution has attracted the attention of statisticians for more than half a century working on theory and methods as well as various fields of statistics.
Azad and Alam (2012) also mentioned that the statisticians has utmost interest in it because of its great number of special features because of its ability to fit to data from various fields, ranging from life data to weather data or observations made in economics and business administration, in health, in physical and social science, in hydrology, in biology or in the engineering sciences.
Akpinar and Akpinar (2004) mentioned that the wind speed probability distributions and the functions representing them mathematically are the main tools used in the wind-related literature. Their use includes a wide range of applications, from the techniques used to identify the parameters of the distribution functions to the use of such functions for analysing wind speed data and wind energy economics.
Akpinar and Akpinar (2004) mentioned that Weibull and Rayleigh's function are two of the commonly used functions for fitting a measured wind speed probability distribution in a given location over a certain period . Mukut (2008) mentioned that the Weibull distribution is characterized by two parameters: the shape parameter k (dimensionless) and the scale parameter c (m/s.) The cumulative distribution function is given by-

$$F(v) = (1 - e)^{-\left(\frac{v}{c}\right)^k} \tag{2.1}$$

And the Weibull density (probability density) function by -

$$f(v) = \frac{dF(v)}{dv} = \left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1} \times e^{-\left(\frac{v}{c}\right)^k} \tag{2.2}$$

Gugliani et al. (2018) mentioned that for v>0 and (k,c)>0, where v is the reference wind speed, k is the non-dimensional shape parameter, and c is the scale parameter whose dimension coincides with that of v(m/s) .
Azad et al. (2014) mentioned that the Weibull density function, which is a three-parameter function, but for wind speed, it can be expressed mathematically in two parameter model .The Weibull Density Function means the relative frequency of wind speeds for the site.Therefore, the shape of wind speeds distribution can be guessed when it is plotted mentioned by Mukut (2008). He again mentioned that Weibull distribution function is the integration of Weibull Density Function. It is the cumulative of relative frequency of each velocity interval.

2.3.1 Weibull Parameter Estimation Methods

To estimate the best value of Weibull parameters a Weibull distribution curve needed to be generated. Chaurasiya et al. (2018) mentioned that the Weibull curve is a probability density function versus wind speed curve generated based on the parameter calculated using nine different numerical methods, giving an idea of the method which fits best to the data of collected wind speed. Furthermore, from the figures, it is possibly easy to verify how the different curve matches the histogram of the measured data.

To calculate parameters of the Weibull distribution there are different numerical methods such as-

(1) Energy Trend Method : Kaplan (2017) mentioned that it is the ratio of total wind energy to the average wind speed. At first we need to calculate energy trend factor (E_{pf}) then shape parameter (k) can be obtained by following equations-

$$E_{pf} = \frac{\frac{1}{n}\sum_{i=1}^{n} v_i^3}{(\frac{1}{n}\sum_{i=1}^{n} v_i)^3} \tag{2.3}$$

$$k = \frac{1}{3.9557 E_{pf}^{0.898}} \tag{2.4}$$

The scale parameter is calculated as follows mentioned by Kaplan (2017)-

$$c = (\frac{1}{n}\sum_{i=1}^{n} v_i^k)^{\frac{1}{k}} \tag{2.5}$$

(2) Moment Method : Usta et al. (2018) mentioned that the estimates of the Moment Method is obtained by equating sample moments with population moments. Thus, the estimates of k and c based on the Moment Method are obtained as a solution of the following equations.

$$\bar{v} = c\,\Gamma(1 + \frac{1}{k}) \tag{2.6}$$

$$\sigma = c\left[\Gamma(1 + \frac{2}{k}) - \Gamma^2(1 + \frac{1}{k})^{0.5}\right] \tag{2.7}$$

(3) Power Density Method : Chaurasiya et al. (2017) mentioned that this is a new and easier method for calculating shape factor and scale factor. This method involve less computation, easier implementation and formulation. The primary step in this method is to compute energy pattern factor E_{pf}, Energy pattern factor is a ratio of mean of cubic wind speed to cube of mean wind speed.

$$E_{pf} = (\frac{\bar{v^3}}{\bar{v}^3}) \tag{2.8}$$

where $\bar{v^3}$ is mean of cube of wind speed and $\bar{v}^3$ is cube of mean speed. $\bar{v}$ can be calculated by-

$$\bar{v} = (\frac{1}{n})\sum_{i=1}^{n} v_i \tag{2.9}$$

After solving above equations Weibull shape parameter k and scale parameter c can be calculated using this formula-

$$k = \left(1 + \frac{3.69}{E_{pf}^2}\right) \tag{2.10}$$

$$c = \frac{\bar{v}}{\Gamma(1+\frac{1}{k})} \tag{2.11}$$

(4) Emperical Method of Justus : The Emperical method is a special case of the Method of Moments to estimate Weibull parameters. Justus et al. (1978) proposed following estimation-

$$k = (\frac{\bar{v}}{\sigma})^{-1.086} \tag{2.12}$$

,

$$c = \left(\frac{\bar{v}}{\Gamma(1+\frac{1}{k})}\right) \tag{2.13}$$

(5) Emperical Method of Lysen : Mohammadi et al. (2016) mentioned that in the empirical method suggested by Lysen, k is calculated same as the Justus method. In fact, the only difference is the equation of c . In the empirical method of Lysen, k and c is obtained by-

$$k = (\frac{\bar{v}}{\sigma})^{-1.086} \tag{2.14}$$

$$c = \bar{v}(0.568 + \frac{0.433}{k}) - \frac{1}{k} \tag{2.15}$$

(6) Maximum Likelihood Method : Mohammadi et al. (2016) mentioned that the maximum likelihood estimation method is a mathematical expression recognized as a likelihood function of the wind speed data in time series format. In this method, extensive numerical iterations are required to determine the k and c parameters of the Weibull function. Using the maximum likelihood method, the k and c parameters are computed by following equations .

$$k = \left(\frac{\sum\limits_{i=1}^{n} v_i^k \, ln v_i}{\sum\limits_{i=1}^{n} v_i^k} - \frac{\sum\limits_{i=1}^{n} ln v_i}{n}\right)^{-1} \tag{2.16}$$

$$c = \left[\frac{1}{n}\sum_{i=1}^{n}(v_i)^k\right]^{\frac{1}{k}} \tag{2.17}$$

(7) Standard Deviation Method : For this we need to calculate the average velocity, v_m and standard deviation of wind, σ_v first mentioned by Bhuiyan et al. (2011) . Equations are given by -

$$V_m = \left(\frac{\frac{1}{n}\sum\limits_{i=1}^{n} f_i v_i}{\sum\limits_{i=1}^{n} f_i}\right)^{\frac{1}{8}} \tag{2.18}$$

$$\sigma_v = \sqrt{\frac{\sum\limits_{i=1}^{n} f_i (v_i - v_m)^2}{\sum\limits_{i=1}^{n} f_i}} \tag{2.19}$$

Once the V_m and σ_v is calculated shape parameter (k) can be calculated numerically by the following equation mentioned by Bhuiyan et al. (2011)-

$$\left(\frac{\sigma_v}{V_m}\right)^2 = \frac{\Gamma(1+\frac{2}{k})}{\Gamma^2(1+\frac{1}{k})} - 1 \tag{2.20}$$

Once k is determined, c is given by-

$$c = \frac{V_m}{\Gamma(1+\frac{1}{k})} \tag{2.21}$$

(8) Alternative Maximum Likelihood Method : Chaurasiya et al. (2017) mentioned that simple calculation procedure has been developed due to iterative characteristics of maximum likelihood method called alternative maximum likelihood method. The weibull parameters can be determined using following equations.

$$k = \frac{\pi}{\sqrt{6}} \left[\frac{n(n-1)}{n(\sum_{i=1}^{n} lnv^2) - \{\sum_{i=1}^{n} lnv\}^2} \right]^{\frac{1}{2}} \tag{2.22}$$

$$c = \left[\frac{1}{n} \sum_{i=1}^{n} (v_i)^k \right]^{\frac{1}{k}} \tag{2.23}$$

(9) Modified Energy Pattern Factor Method: Gugliani et al. (2018) mentioned that for this method we need to calculate E_{pf} from equation **(2.8)**. After that Shape parameter can be calculated from following equation-

$$k = \frac{a_4 E_{pf}^4 + a_3 E_{pf}^3 + a_2 E_{pf}^2 + a_1 E_{pf} + a_o}{b_4 E_{pf}^4 + b_3 E_{pf}^3 + b_2 E_{pf}^2 + b_1 E_{pf} + b_o} \tag{2.24}$$

where a_n and b_n are rational function coefficients. They are determined by using the non linear least squares method with the Lavenberg- Marquardt algorithm. The values of the coefficients are shown below-

a_o = -0.2204	b_o =-1.2728
a_1 = 3.2753	b_1 = 3.6912
a_2 = -5.7896	b_2 = -2.6097
a_3 = 2.1514	b_3 = -0.8005
a_4 = 0.5904	b_4 = 0.9920

Table 2.1: Coefficients for Shape Parameters

Once 'k' is determined , Scale parameter 'c' can be determined from equation **(2.21)** .

2.4 Wind Turbine Types

A wind turbine converts the kinetic energy in wind into mechanical energy. If the mechanical energy is used directly by machineries, such as a pump or grinding stones, the machine is usually called a windmill. If the mechanical energy is then converted to electricity, the machine is called a wind generator.
Wind turbines are classified into two general types: horizontal axis and vertical axis. A horizontal axis machine has its blades rotating on an axis parallel to the ground. A vertical axis machine has its blades rotating on an axis perpendicular to the ground. There are a number of available designs for both and each type has certain advantages and disadvantages. However, compared with the horizontal axis type, very few vertical axis machines are available commercially.

2.4.1 Horizontal Axis Wind Turbine

Saad and Asmuin (2014) mentioned that a horizontal wind turbine has currently been modernized from the traditional windmill designs that have been around for many centuries. A nacelle installed perpendicular to the turbine tower and horizontal in terms of the ground shows the name of the turbine. Most common models for drawing energy from wind, the horizontal wind turbines offer a great amount of advantages. The core components of a horizontal wind turbine are as shown below:

1. The main rotor shaft.
2. The electrical generator
3. The gearbox to increase the rotation speed of the blades.
4. Turbine blades, with stiffness to avoid contacting the post.

A wind vane helps to direct turbines to the wind, while a wind sensor is implemented for a large horizontal wind turbine. Horizontal Axis Wind Turbines have to always be pointed in the right direction so they can have max efficiently. Saad and Asmuin (2014) mentioned that conventional and smaller upwind based turbines, accurate orientation is obtained via the use of a simple wind vane; large turbines include a yaw motor and yaw meter. The yaw meter picks up the accurate direction of the wind, and the yaw motor moves the turbine to ensure it is always positioned into the wind .

2.4.2 Vertical Axis Wind Turbine

Ragheb (2011) mentioned that vertical axis wind turbines are advocated as being capable of catching the wind from all directions and do not need yaw mechanisms, rudders or downwind coning. Their electrical generators can be positioned close to the ground, and hence easily accessible. A disadvantage is that some designs are not self-starting. There have been two distinct types of vertical axis wind turbines: The Darrieus and the Savonius types. The Darrieus rotor was researched and developed extensively by Sandia National Laboratories in the USA in the 1980s. New concepts of vertical axis wind machines are being introduced such as the helical types particularly for use in urban environments where they would be considered safer due to their lower rotational speeds avoiding the risk of blade ejection and since they can catch the wind from all directions.

2.5 Computational Models For HAWT

2.5.1 Blade Element Momentum Theory

Das and Amano (2014) mentioned that typical wind turbine performance analysis is done based on BEM theory. BEM is also referred to as Strip theory. The BEM theory is based on the premise that the forces acting on the wind turbine blades are solely responsible for the change in axial momentum of the air passing through the swept area of the blades.

Ingram (2011) mentioned that Blade Element Momentum Theory equates two methods of examining how a wind turbine operates. The first method is to use a momentum balance on a rotating annular stream tube passing through a turbine. The second is to examine the forces generated by the airfoil lift and drag coefficients at various sections along the blade. These two methods then give a series of equations that can be solved iteratively .

2.5.2 Blade Element Theory

Ingram (2011) mentioned that Blade element theory relies on two key assumptions:

- There are no aerodynamic interactions between different blade elements
- The forces on the blade elements are solely determined by the lift and drag coefficients.

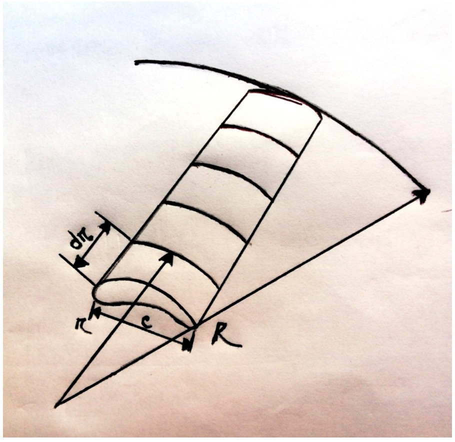

Figure 2.2: Rotating Annular Stream tube: notation

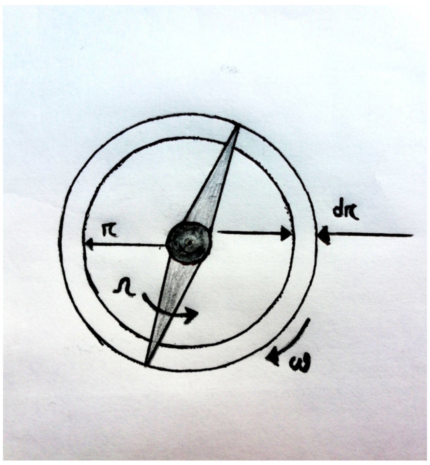

Figure 2.3: The Blade Element Model

Consider a blade divided up into N elements as shown in Figure (2.2). Each of the blade elements will experience a slightly different flow as they have a different rotational speed (ωr) (Fig 2.3), a different chord length (c) and a different twist angle (γ). Blade element theory involves dividing up the blade into a sufficient number (usually between ten and twenty) of elements and calculating the flow at each one. Overall performance characteristics are determined by numerical integration along the blade span.

2.5.3 Momentum Theory

Das and Amano (2014) mentioned that Momentum theory refers to the control volume analysis of the forces at the blade based on the conservation of linear and angular momentum, whereas blade element theory refers to the analysis of forces at a section of the blade, as a function of blade geometry.

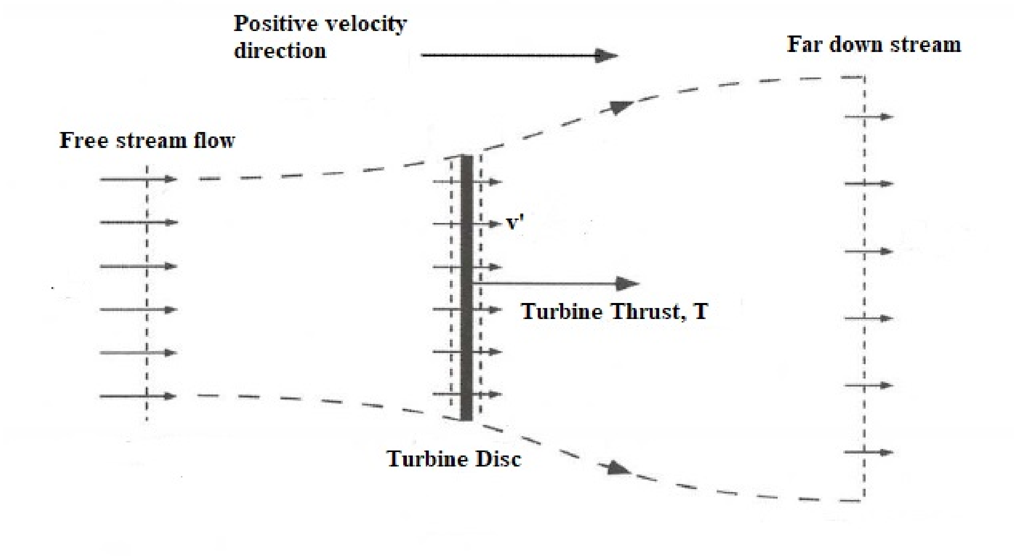

Figure 2.4: Actuator disc and stream tube

Das and Amano (2014) mentioned that as mechanical energy can only be extracted from a change in kinetic energy and hence velocity of the wind stream, the wind velocity behind the wind turbine has to be less than the wind velocity in front of the wind turbine. This would imply an increase in area since the density change is insignificant. Utilizing an actuator disc and stream-tube assumption as shown in Fig (2.4), we can estimate the power conversion from a wind turbine.
The mechanical energy that is extracted can be found by subtracting the available kinetic energy after the actuator disc from the initial free stream kinetic energy .

$$P = \frac{1}{2}\rho(v_1^3 A_1 - v_2^3 A_2) \tag{2.25}$$

By continuity equation,

$$\rho v_1 A_1 = \rho v_2 A_2 \tag{2.26}$$

Thus, eq **(2.26)** yields as-

$$P = \frac{1}{2}\rho v_1 A_1(v_1^2 - v_2^2) \tag{2.27}$$

$$P = \frac{1}{2}\dot{m}(v_1^2 - v_2^2) \tag{2.28}$$

As per eq **(2.28)**, for power extracted to be maximum, v_2 has to be zero. But this is physically impossible since it would imply no flow across the turbine mentioned by Das and Amano (2014).
Using conservation of momentum, the force exerted by the air on the disc can be expressed as-

$$\bar{F} = \dot{m}(v_1 - v_2) \tag{2.29}$$

Assuming that the velocity of air in the disc plane to be v', the power required for moving the air at this velocity from the disc plane is-

$$P = \bar{F}v' = \dot{m}(v_1 - v_2)v' \tag{2.30}$$

From equation **(2.29)** and **(2.30)** we get-

$$v' = \frac{1}{2}(v_1 + v_2) \tag{2.31}$$

Hence, $\dot{m} = \frac{1}{2}\rho A(v_1 + v_2)$ (2.32)

And mechanical power output from the disc-

$$P = \frac{1}{4}\rho A(v_1 + v_2)(v_1^2 - v_2^2) \tag{2.33}$$

The coefficient of power C_p is defined as the ratio of extracted power to the available power.

$$C_p = \frac{P}{P_o} = \frac{\frac{1}{4}\rho A(v_1 + v_2)(v_1^2 - v_2^2)}{\frac{1}{2}\rho A v_1^3} \tag{2.34}$$

By rearranging the terms,

$$C_p = \frac{P}{P_o} = \frac{1}{2}\left[1 - (\frac{v_2}{v_1})^2\right]\left[1 + \frac{v_2}{v_1}\right] \tag{2.35}$$

The induction factor a is defined as-

$$a = \frac{v_2}{v_1} \tag{2.36}$$

The torque coefficient is estimated as Hansen (2015)-

$$C_T = \frac{ThrustForce}{DynamicForce} = 4a(1 - a) \tag{2.37}$$

2.5.4 Betz's Law

Wagner (2013) mentioned that we can only convert a maximum of 59$\%$ of the kinetic energy in the wind to mechanical energy using a wind turbine which is known as Betz's Law. This is because the wind on the back side of the rotor must have a high enough velocity to move away and allow more wind through the plane of the rotor.

Das and Amano (2014) mentioned that for maximum power extraction, $dcp/d(v2/v1)$ has to be zero, which implies for maximum power output -

$$\frac{v_2}{v_1} = \frac{1}{3} \tag{2.38}$$

and,

$$C_p = \frac{16}{27} = 0.593 \tag{2.39}$$

2.6 Computational Tools

For design and simulation, it is important to choose the right computational tool for the right operation. Here are some computational tools mentioned below-

- Matlab
- CFD
- OpenFOAM
- QBlade
- PSCAD

2.6.1 QBlade

For developing a HAWT wind turbine blade it is necessary to select performance and design analysis software. For this researchers are using ANSYS, Matlab and other high-performance computing software's. These software's require high-performance computers and resources. Moreover, they are much time consuming and not available to use it free. That's why for this project QBlade (2016) is chosen to design and simulate HAWT because it's free and requires less time to work. QBlade which is an open-source, cross-platform simulation software for wind turbine blade design and aerodynamic simulation. It was developed in TU Berlin which is a reputed University of Germany, Marten and Wendler (2013).
Parikh (2018) mentioned that QBlade makes it easy to understand the blade performance by plotting power and load curves in an easy and intuitive way. With QBlade we can also conduct rotor simulations and it gives good understanding with rotor variables and geometry .

2.7 Design Parameters Related To HAWT

Design parameters are obligatory elements for constructing a well performing HAWT blade. Some elements have much more importance than other parameters such as blade airfoil, tip speed ratio, rated wind etc for getting the best output from the designed blade. Table 2.2 shows different important design parameters, Islam et al. (2011).

Category	Parameters
Physical Features	1. Airfoil 2. Number of Blade 3. Angle of Attack
Dimensional	1. Swept Area 2. Solidity
Operational	1. Tip Speed Ratio 2. Power Coefficient 3. Rated Speed 4. Cut-in Speed 5. Cut-off Speed
Balance System	1. Tower 2. Breaking Mechanism
Others	1. Materials 2. Aesthetics

Table 2.2: Different Design Parameters of a HAWT

2.7.1 Airfoils

The cross-sectional profile that is obtained by the intersection of an airplane wing, blade of a rotor or turbine with a perpendicular plane is known as an airfoil, Islam et al. (2015). There are different shapes and sizes of airfoil depending on the specifications and configuration. Different types of Airfoil are- Symmetric type, Asymmetric type, Flat bottom type, etc. Asymmetric airfoils can be both positive cambered and negative cambered. In aircraft and wind turbine blade, mostly asymmetric airfoils are used because it creates a good amount of lift even at small AOA and it has improved lift to drag ratio and better stall characteristics which are great advantages. Asymmetrical airfoils have disadvantages too. The disadvantages are center of pressure travel of up to 20 percent of the chord line (creating undesirable torque on the airfoil structure) and greater production costs. On the other hand, symmetrical airfoils are unable to produce any lift at zero AOA but it can produce an equal amount of lift in either direction at the same positive or negative AOA.

2.7.2 Number of Blades

Mühle et al. (2016) mentioned that in the wind power industry the development and research in the last decades focused mostly on 3-bladed turbines. Whereas in the 1970's and 1980's 2-bladed turbines were still investigated and considered in turbine development, the research effort in 2-bladed turbines was only minor in the last years. This is due to the disadvantages of 2-bladed rotors compared to 3-bladed rotors, such as the higher noise emissions, the distracting visual effects and the unfavorable dynamic behavior. However, as the offshore wind energy market is gaining importance, the 2-bladed turbines are getting more significant again. This is due to the fact, that the drawbacks are not so much relevant offshore and the big advantage of one less rotor is strongly decreasing the costs. Nevertheless the maximum theoretical attainable performance is increasing with increasing blades number because of reduced tip vortices, which is also a drawback for 2-bladed rotors when comparing them to the established 3-bladed rotors. This fact should also be considered when looking at the economic potential of the rotor blade number. Moreover, in a wind farm set-up, wake effects are especially of interest to evaluate how the turbines interact and hence, their implication on wind farm design and power production .

2.7.3 Tip Speed Ratio

Tip Speed Ratio (often known as the TSR) is an important function in the design of wind turbine. If the rotor of the wind turbine turns too slowly, most of the wind will pass through the gap between the rotor blades and winds will remain undisturbed. Alternatively, if the rotor turns too quickly, the blades will appear like a solid wall to the wind. In this case, all wind will pass over the blades. Therefore, wind turbines are designed with optimal tip speed ratios to extract as much power out of the wind as possible. The turbulence of wind is generated in the wake region of the wind turbine when a rotor blade passes through the air. If the next blade on the spinning rotor arrives at this point while the air is still turbulent, it will not be able to extract power efficiently from the wind. Therefore the tip speed ratio is chosen so that the blades do not pass through too much turbulent air. Tip speed ratio can be denoted as -

$$TSR = \frac{\omega r}{v_\infty} \tag{2.40}$$

2.7.4 Operational Wind Speed

- **Cut-in Speed -** At very low wind speeds, there is insufficient torque exerted by the wind on the turbine blades to make them rotate. However, as the speed increases, the wind turbine will begin to rotate and generate electrical power. The speed at which the turbine first starts to rotate and generate power is called the cut-in speed. In this research for 3kW HAWT cut-in speed is set to 2m/s for the simulation.

- **Rated Speed -** As the wind speed rises above the cut-in speed, the power output reaches the limit that the electrical generator is capable of. This limit to the generator output is called the rated power output and the wind speed at which it is reached is called the rated output wind speed. The rated wind speed is set to 10m/s.

- **Cut-out Speed -** As the speed increases above the rate output wind speed, the forces on the turbine structure continue to rise and, at some point, there is a risk of damage to the rotor. As a result, a braking system is employed to bring the rotor to a standstill. This is called the cut-out speed. Here cut-out speed is set to 16 m/s for 3kW HAWT.

2.7.5 Angle of Attack

Sestito et al. (2018) mentioned that Angle of attack is the angle between the chord line of an airfoil and the relative wind flow. The greater the angle of attack, the more lift is generated by the wing but it has a limit limit. As long as air flowing smoothly over airfoil it will generate lift but at higher AOA the lift generated drops dramatically and causes stall.
Le Vie (2014) mentioned that this happens because the flow over the airfoil gets disturbed at high AOA and doesn't streems smoothly over airfoil. This is also known as Critical AOA . Figure 2.5 illustrates the AOA-

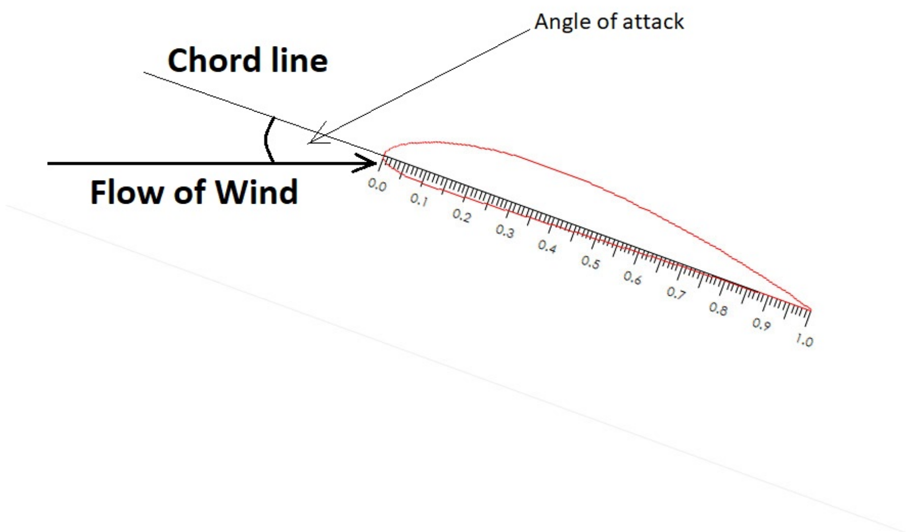

Figure 2.5: Angle of Attack

2.7.6 Power Coefficient, Cp

For wind turbine design the most important parameter is Cp also known as power coefficient.
Wood (2011) mentioned that Cp is the ratio of the actual power produced to the power in the wind what would pass through the blade.

$$C_p = \frac{P}{\frac{1}{2}\rho A V^3} \tag{2.41}$$

The value of $C_{p,max}$ can be expressed from Betz's limit.

Chapter 3

METHODOLOGY

3.1 Flow Process of The Research

Figure 3.1 illustrates flow diagram of our working procedure to reach our objectives.

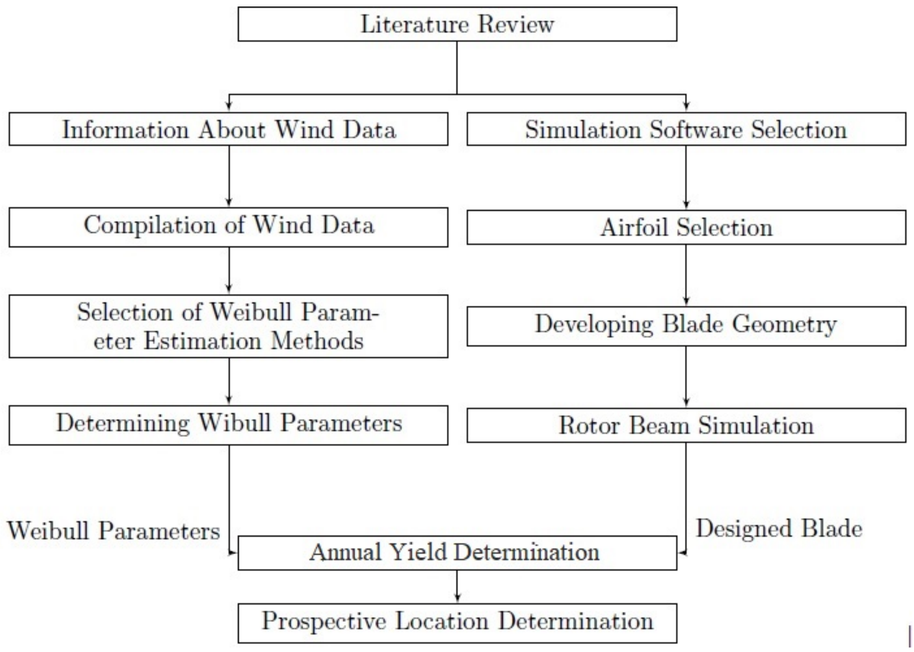

Figure 3.1: Flow Process of research

3.2 Compilation of Wind Data

For the research, wind data at 40m height has been collected for nine meteorological stations of Bangladesh. These nine meteorological stations are- Rangpur, Rajshahi, Mymensingh, Mirzapur(Sylhet), Mongla, Chandpur, Shitakunda, Parkay Beach. These nine meteorological stations installed through NREL's Bangladesh Wind Resource Mapping Project were used to collect measurement data from June 2014 to December 2017. The observation site installation reports, raw and processed data sets, and modeled wind resource data products can be downloaded from instructions found at- (https://www.re-explorer.org/bangladesh-data.html)

3.3 Selection of Weibull Parameters Estimation Methods

After collecting all necessary wind data for prospective locations of Bangladesh, it is needed to select suitable Weibull parameter estimations methods to determine Weibull shape and scale parameters. To calculate parameters of the Weibull distribution four different numerical methods are used in this study. They are-

1. Power Density Method
2. Alternative Maximum Likelihood Method
3. Standard Deviation Method
4. Modified Energy Pattern Factor Method

These methods have been already discussed briefly in Literature Review section.

3.3.1 Determination of Weibull Parameters through Different Methods using Python

Python (1991) is a high level, elucidate open-source programming language with an easy syntax. We choose python because its language constructs an approach and aim to help programmers write clear, logical code for small and large-scale projects. For plotting, we used matplotlib, pandas and seaborn library. We used numpy library to use Gamma, ln and Pi function. After that conventional coding approach was taken to calculate the k and c values. After calculating k and c values PDF was calculated for every method. Then by calling "plot-histogram( )" function we plotted the PDF Vs Wind Speed curve and plotted the best fit curve. As our file was ".csv" type we used python's csv reader to read the data and read the wind speed data and convert them to float data prior to calculation. We calculated Sigma, v^3, E_{pf}, prior to our calculation as well.

3.4 Selection of Airfoil

To design a HAWT blade, airfoil selection is an important part of getting the best performing wind turbine blade. In this research 3m, rotor diameter HAWT blade is designed using different airfoils. At first, NREL's proposed S-series airfoils such as S822, S823, S833, S834, S835 were used and analysis has been made. Later NACA-series airfoils such as NACA 2818, NACA 4818, NACA 4415, NACA 4615 were used for designing new turbine blades for further analysis .

3.4.1 NREL’s Proposed S-Series Airfoils

Figure 3.2 illustrates the selected S-series airfoils-

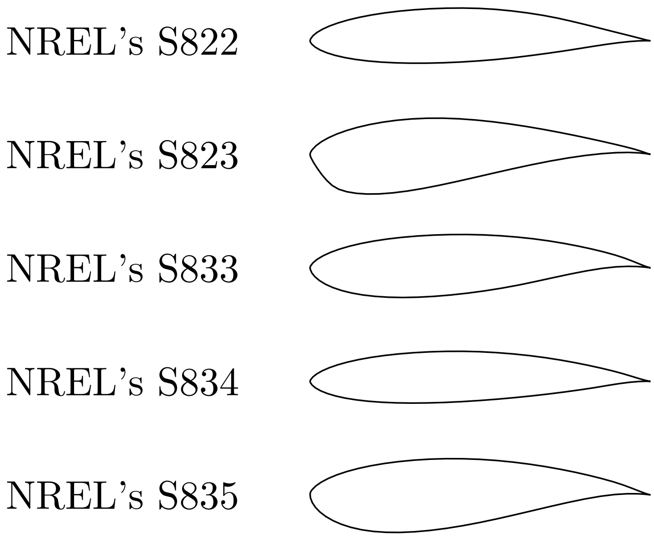

Figure 3.2: NREL’s S series airfoils

3.4.2 NACA Series Airfoils

Figure 3.3 illustrates the selected NACA-series airfoils-

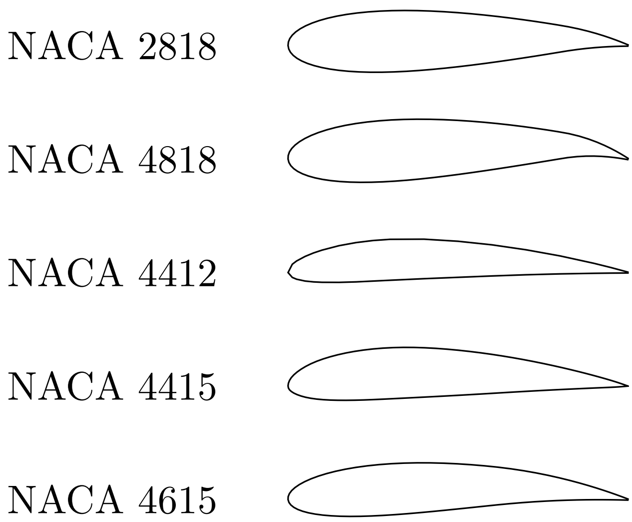

Figure 3.3: NACA series airfoils

3.5 Blade Geometry

After the selection of airfoils, the blade design was developed using QBlade. Blades airfoil position, chord lengths, and twists data was used from the experiment of Anderson et al. (1982).

It should be considered that different blade elements are encountering different magnitudes of Reynolds numbers based on variable relative wind speed and induction factors (axial and radial) as outlined in the QBlade manual. But in this research, a simpler approach has been adopted. A Range of Reynolds number has been applied for all the blade elements. Table 3.1 illustrates the data obtained from the Anderson et al. (1982).

Element	Position(m)	Chord(m)	Twist(°)
1	0.133	0.250	24.21
2	0.233	0.232	21.06
3	0.297	0.199	15.87
4	0.363	0.173	11.92
5	0.427	0.151	8.96
6	0.493	0.133	6.77
7	0.552	0.119	5.18
8	0.623	0.106	4.02
9	0.687	0.097	3.16
10	0.752	0.089	2.49
11	0.818	0.083	1.93
12	0.883	0.077	1.44
13	0.947	0.072	0.99
14	1.012	0.068	0.59
15	1.077	0.064	0.25
16	1.143	0.060	-0.06
17	1.208	0.057	-0.36
18	1.273	0.054	-0.67
19	1.337	0.051	-0.98
20	1.403	0.048	-1.28
21	1.467	0.047	-1.59
22	1.500	0.046	-1.74

Table 3.1: Blade Geometry of Anderson et al's Work

3.6 Data Validation

Validation is described as the degree to which a research study measures what it intends to measure. The results of a study are meaningless if they are not validated properly. If it does not measure what is wanted then the results cannot be used to answer the research question, which is the main aim of the study. These results cannot then be used to generalize any findings and become a waste of time and effort.
To validate the results obtained from any computational tool, there is a need for experimental data. For this research, the valuable work conducted by Anderson et al. (1982) has been considered.
Anderson et al.(1982) carried out a wind tunnel test using a 3m rotor diameter wind turbine blade with NACA 4412 airfoil at wind speed of 10 m/s. Anderson et al.(1982) developed a Cp VS TSR curve which gives peak Cp value of 0.45 at TSR 10. We aimed to develop similar graph closer to the value of Anderson et al.(1982) Cp vs TSR graph with Qblade software using same blade geometry shown in section 3.5 using the same NACA 4412 airfoil to validate our work.

3.7 Development of New Blade of NREL's Proposed Airfoils

In this section, we will discuss the development of 2types of blade design according to the suggested airfoils by NREL for 3m rotor diameter. (National Renewable Energy Laboratory). Table (3.2) illustrates the suggested airfoils for different position of the blade.

Rotor Diameter	Category	Root	Primary	Tip
1-3 m	Thick	S835	S833	S834
3-10 m	Thick	S823	-	S822

Table 3.2: NREL's S Series Airfoils (https://wind.nrel.gov/airfoils/AirfoilFamilies.html)

As we are focused on developing wind turbine blade having a rotor diameter of 3meter, NREL suggested to use S835, S833, S834 airfoils for the rotor diameter of 1-3meter, and for 3-10meter NREL suggested using S823, S833, S822 airfoils. As both suggested option includes 3meter rotor diameter, so we decided to develop two types of blade design according to it and analyze their performance.

3.8 Development of New Blade With NACA Series Airfoils

We also designed blades with NACA series airfoils which are NACA 2818, NACA 4818, NACA 4415, NACA 4615. 4 blades were designed using the same airfoil in every section(top, mid and root) of the blade and same blade geometry was used that has been mentioned in section 3.5. After that their performance was evaluated.

3.9 Annual Yield Determination

After choosing the best performing wind turbine blade design, the Annual Yield of power generation was quantified using the calculated Weibull shape (k) and scale (c) parameters. Besides "Turbine BEM

Simulation" sub module from Qblade software was also used with shape and scale parameter values. According to the turbine BEM simulation submodule, a turbine can only be defined if a blade design is already stored in the database, Marten and Wendler (2013). The user needs to specify all turbine parameters and save the designed turbine in the QBlade software. Then a turbine simulation over a range of wind speeds is conducted. When the k and c values for the Weibull distribution are given, the annual yield is computed automatically.

Chapter 4

RESULTS & DISCUSSION

4.1 Shape and Scale Parameters for Prospective Locations

The shape and scale parameters of Weibull Distribution for nine prospective locations has been calculated iteratively solving the four selected methods discussed in the section (2.3.1) using the real wind data collected from the RE Explorer at 40m height. Table (4.1-4.9) showing the result-

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	1.91	4.09
Alternative Maximum Likelihood Method	1.31	3.78
Standard Deviation Method	1.98	4.09
Modified Energy Pattern Factor Method	1.90	4.09

Table 4.1: Shape and Scale Parameters of Chandpur

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.18	5.32
Alternative Maximum Likelihood Method	1.42	4.92
Standard Deviation Method	2.28	5.31
Modified Energy Pattern Factor Method	2.17	3.32

Table 4.2: Shape and Scale Parameters of Inani

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	1.94	3.91
Alternative Maximum Likelihood Method	1.29	3.61
Standard Deviation Method	1.98	3.91
Modified Energy Pattern Factor Method	1.92	3.91

Table 4.3: Shape and Scale Parameters of Mirzapur

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.14	4.59
Alternative Maximum Likelihood Method	1.39	4.21
Standard Deviation Method	2.20	4.58
Modified Energy Pattern Factor Method	2.13	4.58

Table 4.4: Shape and Scale Parameters of Mongla

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.00	3.64
Alternative Maximum Likelihood Method	1.28	3.35
Standard Deviation Method	2.03	3.63
Modified Energy Pattern Factor Method	2.00	3.64

Table 4.5: Shape and Scale Parameters of Mymensingh

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.17	5.36
Alternative Maximum Likelihood Method	1.44	4.96
Standard Deviation Method	2.30	5.35
Modified Energy Pattern Factor Method	2.16	5.36

Table 4.6: Shape and Scale Parameters of Parkay Beach

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.00	3.64
Alternative Maximum Likelihood Method	1.28	3.35
Standard Deviation Method	2.03	3.64
Modified Energy Pattern Factor Method	2.00	3.64

Table 4.7: Shape and Scale Parameters of Rajshahi

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	2.00	3.64
Alternative Maximum Likelihood Method	1.28	3.48
Standard Deviation Method	2.03	3.64
Modified Energy Pattern Factor Method	2.00	3.63

Table 4.8: Shape and Scale Parameters of Rangpur

Method	Shape Factor(k)	Scale Factor(c)
Power Density Method	1.72	3.93
Alternative Maximum Likelihood Method	1.29	3.69
Standard Deviation Method	1.73	3.93
Modified Energy Pattern Factor Method	1.72	3.93

Table 4.9: Shape and Scale Parameters of Shitakunda

The Weibull k value, or Weibull shape factor, is a parameter that reflects the breadth of distribution of wind speeds. A small value for k signifies very variable winds, while constant winds are characterized by a larger k. Weibull scale parameter is signified by c, in m/s and it defines the characteristics of wind speed distribution. c is proportional to the mean wind speed, $\bar{x}$.

4.2 PDF Vs Wind Speed Distribution Curves

The following curves in figure (4.1-4.9) represent the PDF Vs Wind Speed Distribution for four selected methods. They are- PDM, AMLM, SDM, and MEPFM. These curves show the average distribution of wind speeds in a certain area with the probability of getting that specific wind speed.

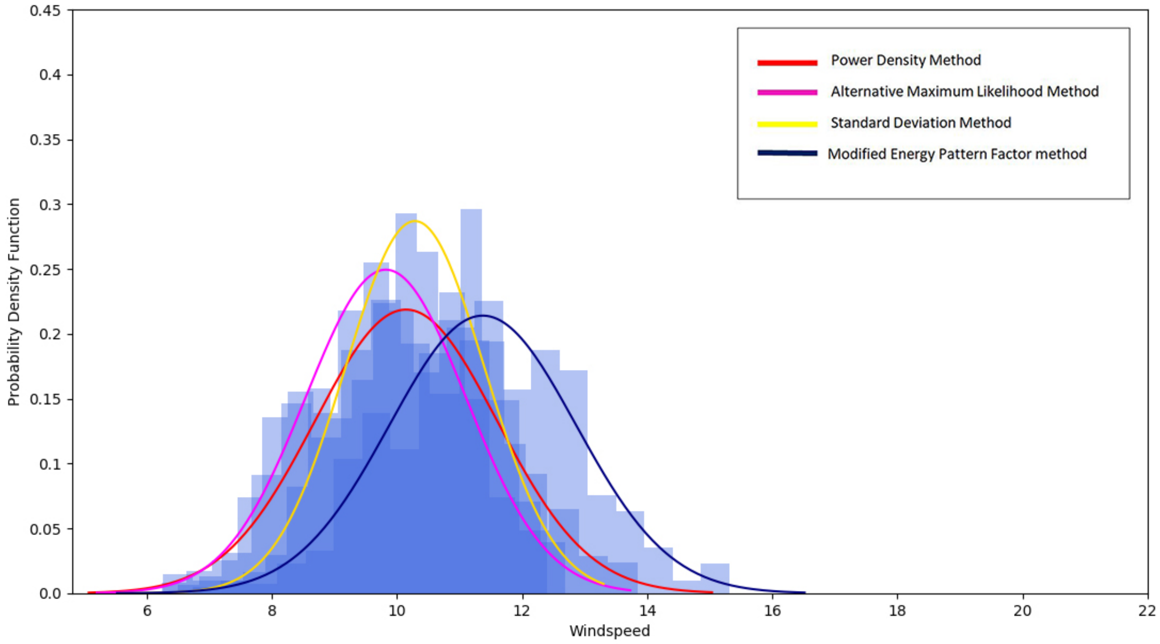

Figure 4.1: PDF Vs Wind speed curve for Chandpur

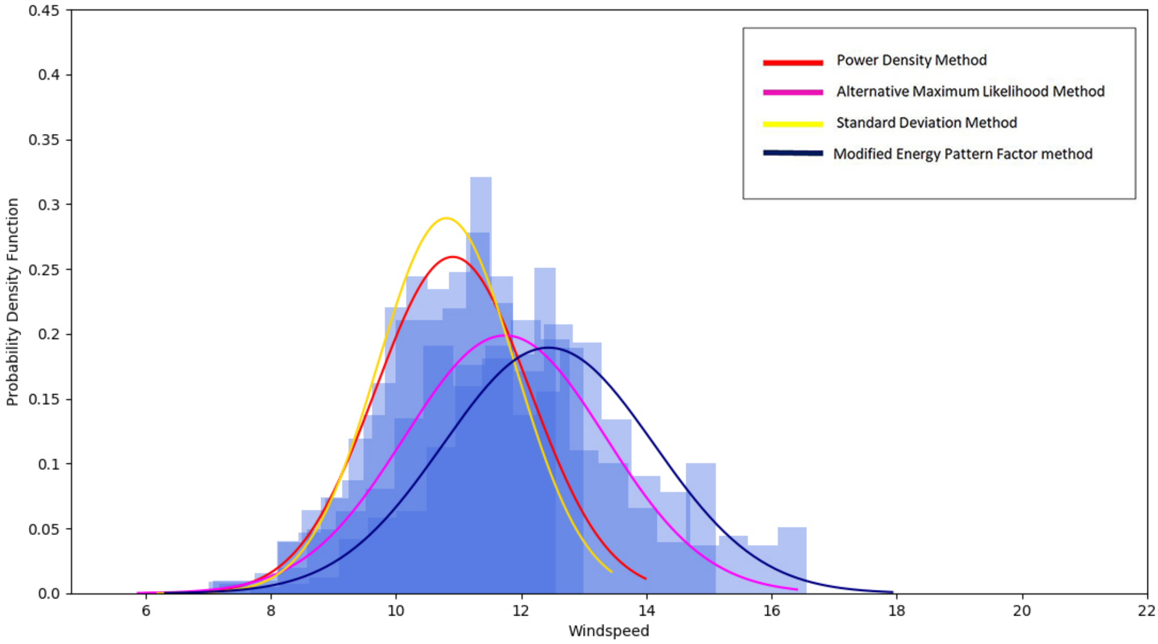

Figure 4.2: PDF Vs Wind speed curve for Inani

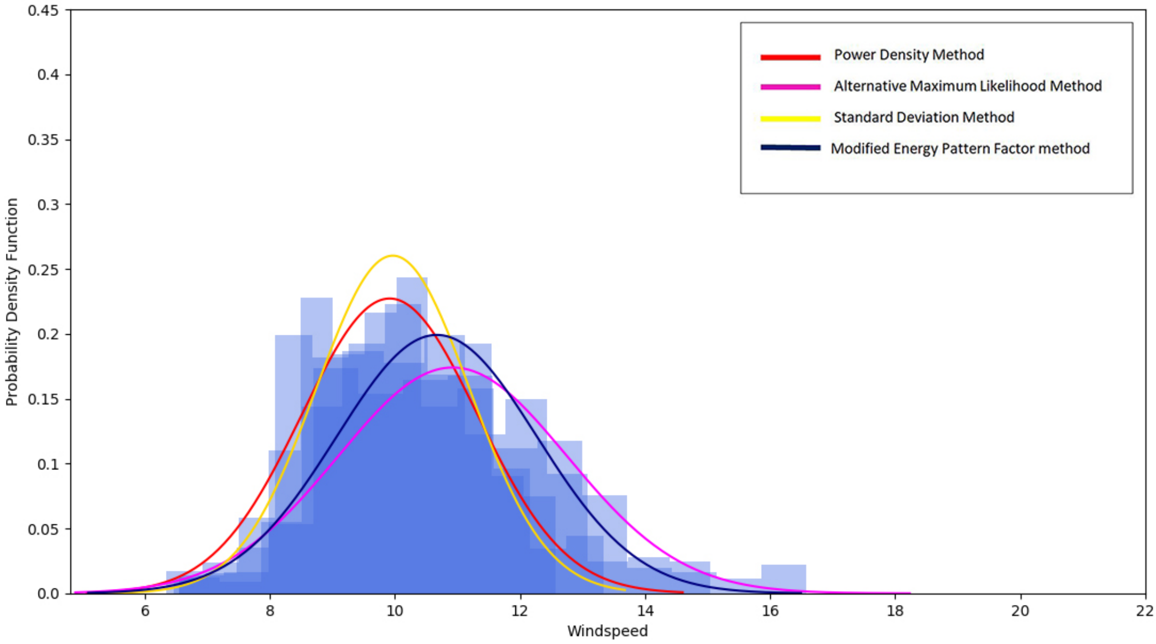

Figure 4.3: PDF Vs Wind speed curve for Mirzapur

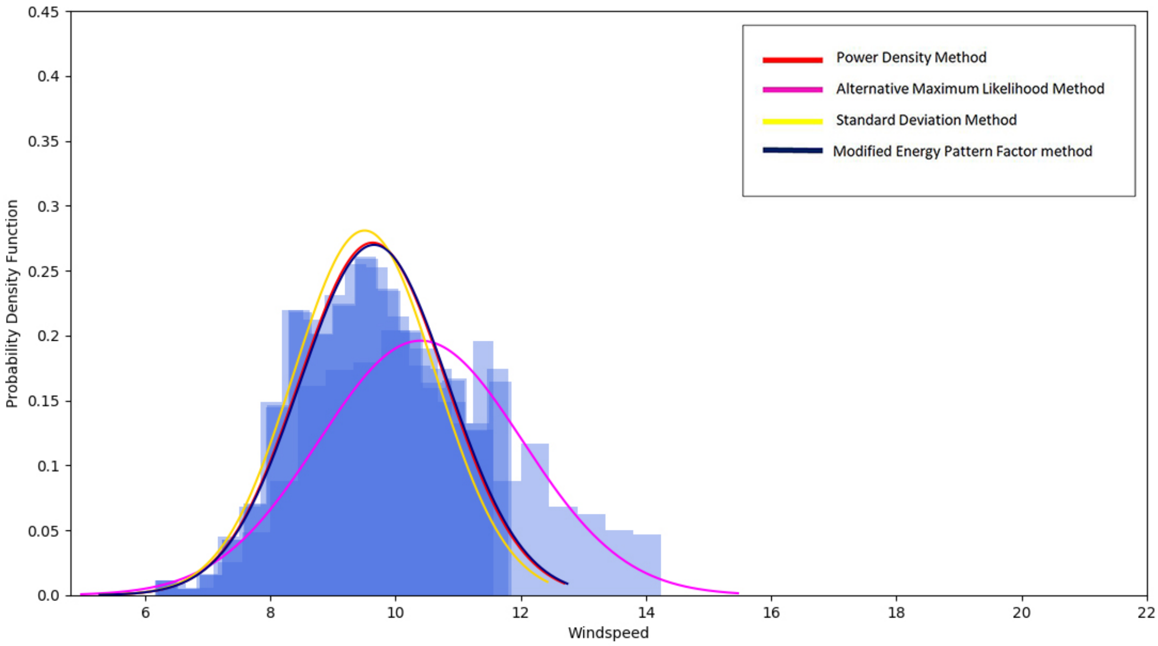

Figure 4.4: PDF Vs Wind speed curve for Mongla

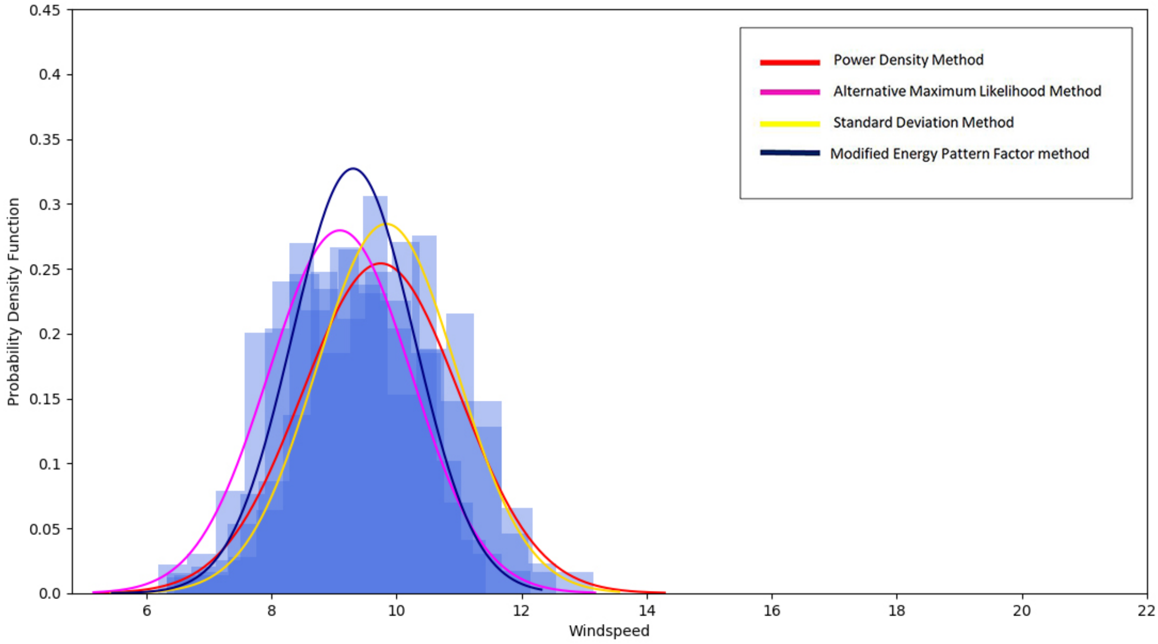

Figure 4.5: PDF Vs Wind speed curve for Mymensingh

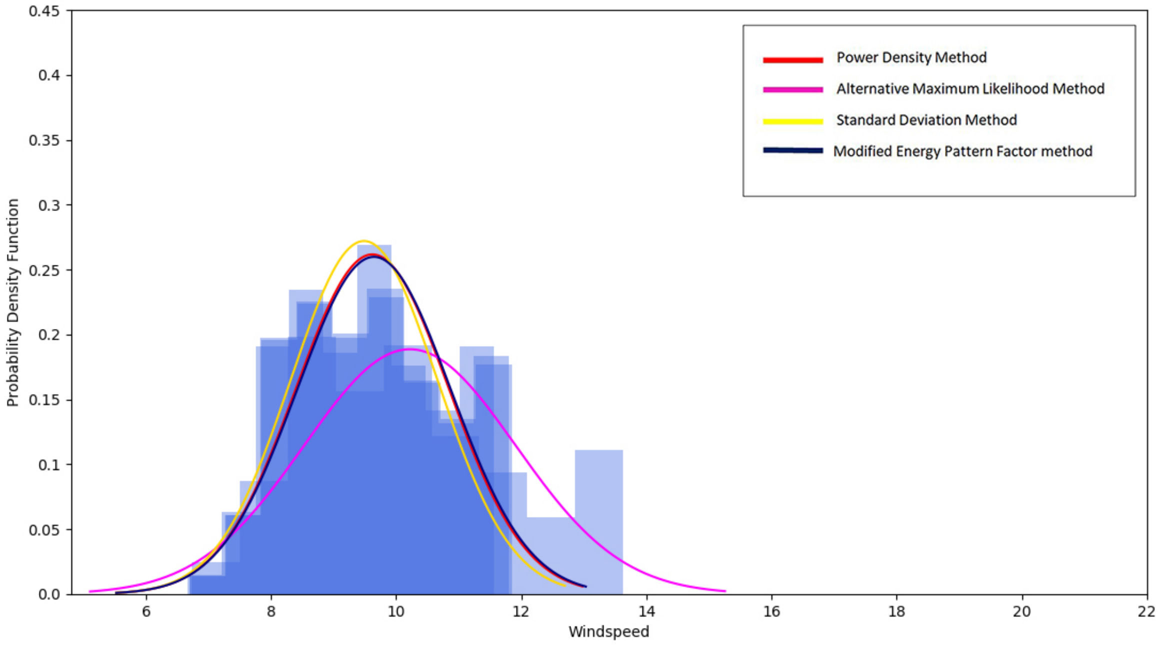

Figure 4.6: PDF Vs Wind speed curve for Parkay Beach

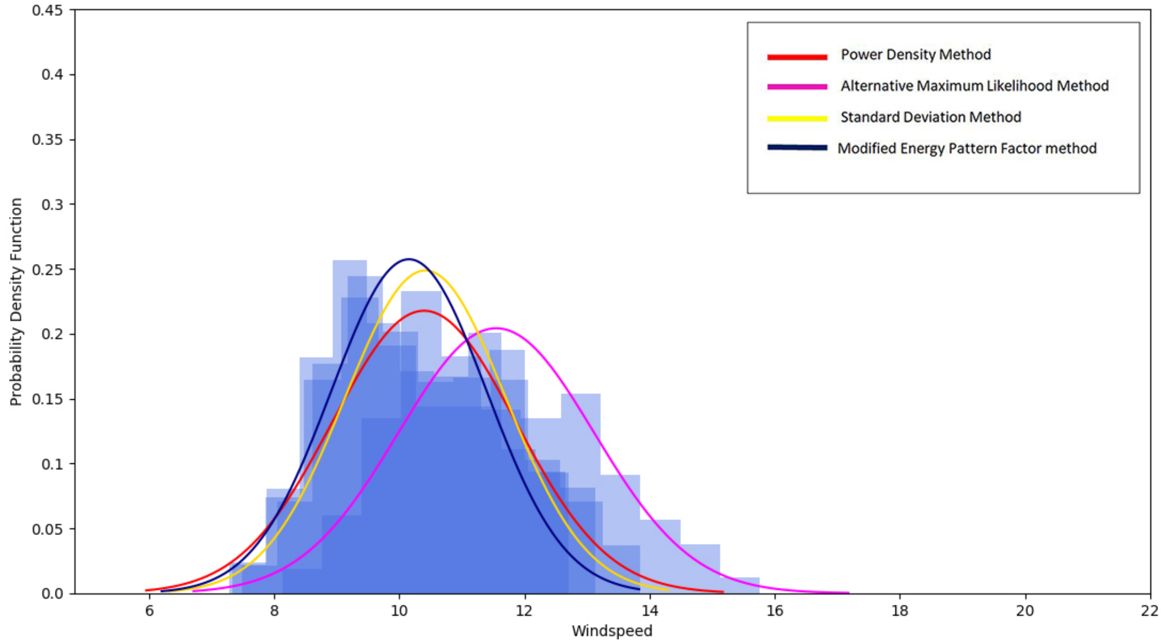

Figure 4.7: PDF Vs Wind speed curve for Rangpur

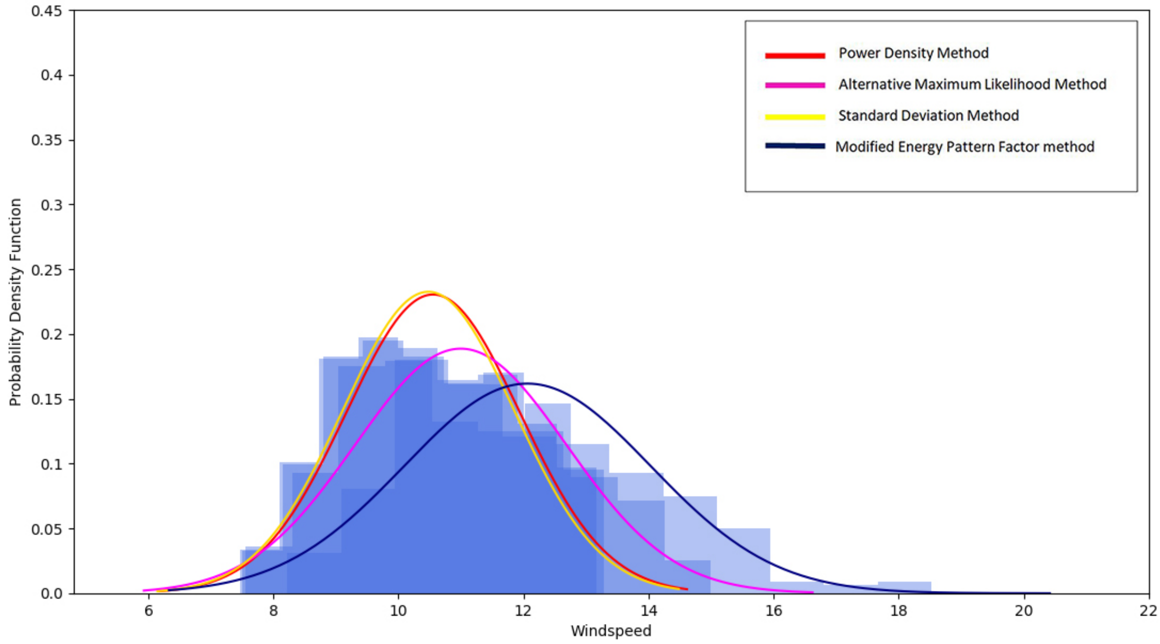

Figure 4.8: PDF Vs Wind speed curve for Rajshahi

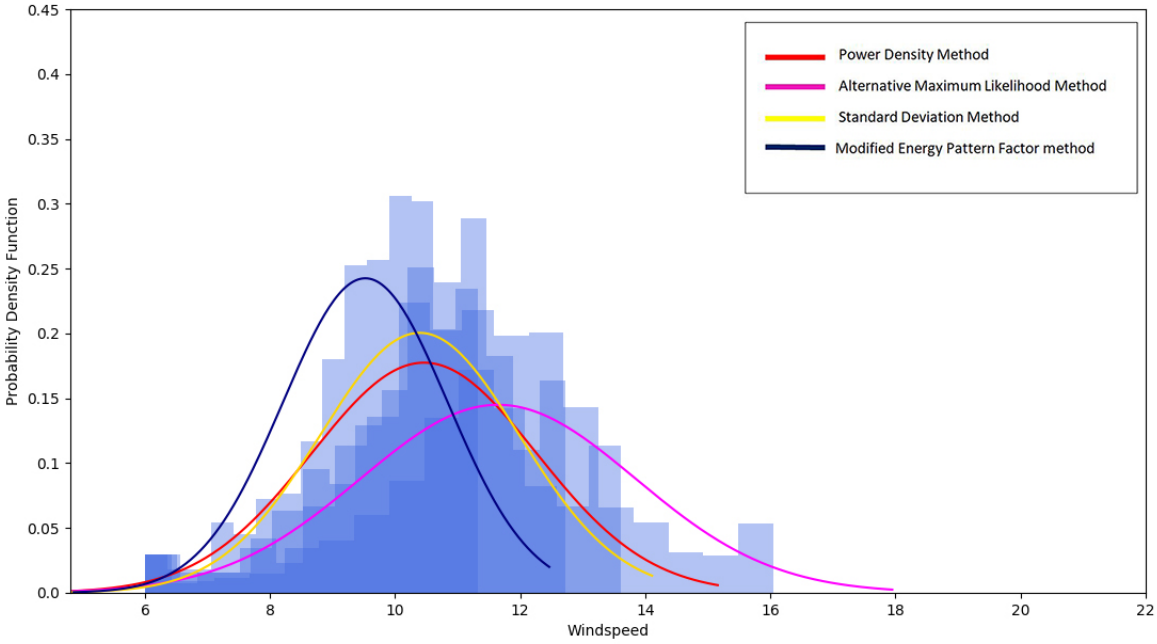

Figure 4.9: PDF Vs Wind speed curve for Shitakunda

4.3 Designed Blade For Validation

The importance of data validation is already mentioned in section (3.6). For our computational software QBlade's result validation, we chose Anderson et al. (1982) experimental work. NACA 4412 airfoil was used for the blade design. Figure (4.12) illustrates the validation result of the Cp vs TSR Curve between our computational result and Anderson et al's experimental result. The blade geometry which has been used for the blade is shown in table (4.10) and NACA 4412 Airfoil profile are shown in figure (4.10) and our computational blade that has been designed with these configurations are shown in figure (4.11)-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	NACA4412
2	0.233	0.232	21.06	NACA4412
3	0.297	0.199	15.87	NACA4412
4	0.363	0.173	11.92	NACA4412
5	0.427	0.151	8.96	NACA4412
6	0.493	0.133	6.77	NACA4412
7	0.552	0.119	5.18	NACA4412
8	0.623	0.106	4.02	NACA4412
9	0.687	0.097	3.16	NACA4412
10	0.752	0.089	2.49	NACA4412
11	0.818	0.083	1.93	NACA4412
12	0.883	0.077	1.44	NACA4412
13	0.947	0.072	0.99	NACA4412
14	1.012	0.068	0.59	NACA4412
15	1.077	0.064	0.25	NACA4412
16	1.143	0.060	-0.06	NACA4412
17	1.208	0.057	-0.36	NACA4412
18	1.273	0.054	-0.67	NACA4412
19	1.337	0.051	-0.98	NACA4412
20	1.403	0.048	-1.28	NACA4412
21	1.467	0.047	-1.59	NACA4412
22	1.500	0.046	-1.74	NACA4412

Table 4.10: Geometry of Bade Using NACA 4412 Airfoil For Data Validation

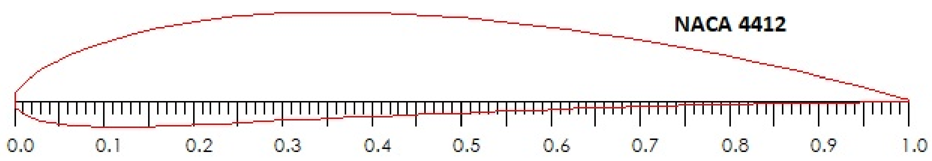

Figure 4.10: NACA 4412 Airfoil

Figule (4.11) illustrates the designed blade using NACA 4412 for validation-

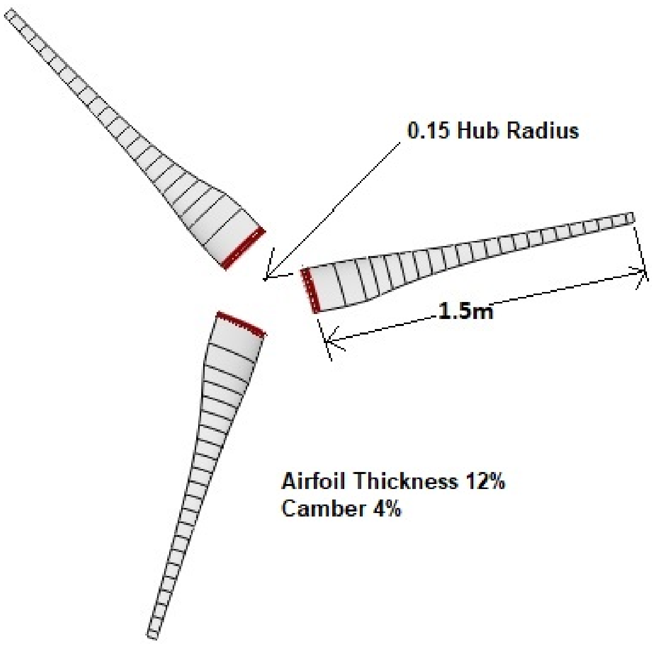

Figure 4.11: NACA 4412 Blade design

4.4 Validation Result

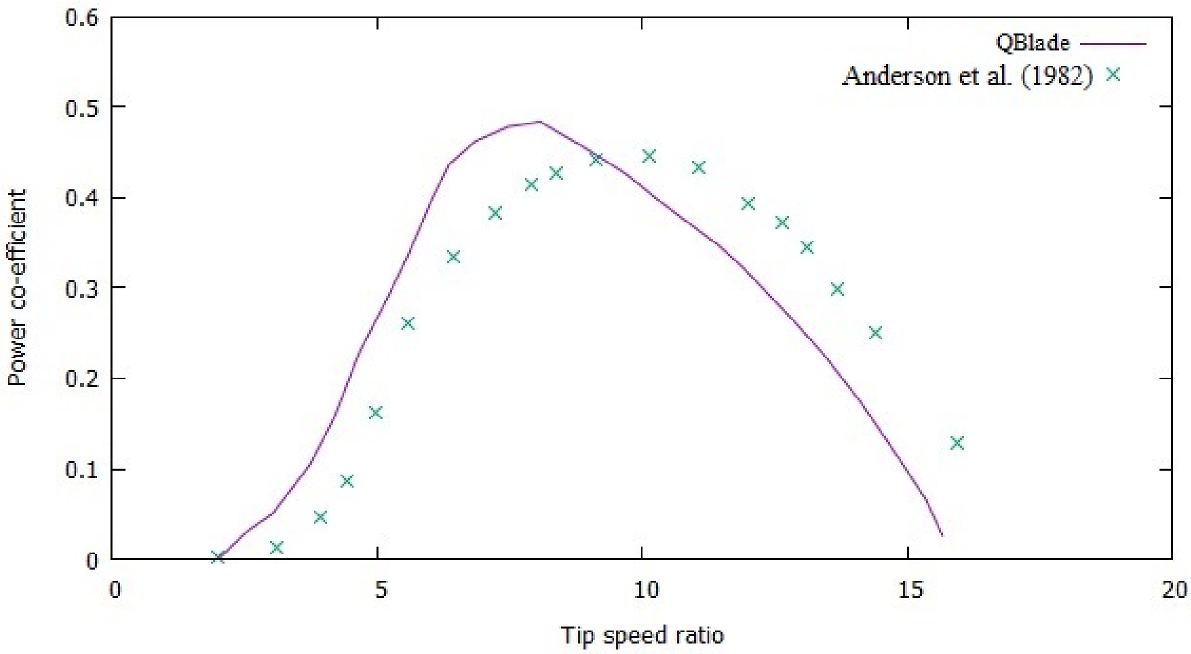

Figure 4.12: Cp vs TSR validation graph

We aimed to generate a similar graph compared to the Anderson at al.(1982) experimental result with the data that we compiled. Here we can see that our computational graph result is conforming with the experimental graph. So we can use the data for further analysis.

4.5 New Blade Design

In this section, we will discuss the newly designed blades for our research. We aimed to design a better performing blade compared to the Anderson et al.(1982) designed turbine. For this, at first, we designed two types of blade for a rotor diameter of 3m proposed by NREL and also designed more blades using NACA series airfoils and compared their results.

4.5.1 NREL's Design Type 1 For Rotor Diameter 1-3 meter

According to the section (3.7), NREL suggests using S835 airfoil at root section of the airfoil, S833 at primary or midsection and S834 at the tip. Using the same geometry as shown in section (3.5) and with proposed airfoils, the blade design was developed. Table (4.11) and figure (4.13) illustrates the blade geometry and design accordingly-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	S835
2	0.233	0.232	21.06	S835
3	0.297	0.199	15.87	S835
4	0.363	0.173	11.92	S833
5	0.427	0.151	8.96	S833
6	0.493	0.133	6.77	S833
7	0.552	0.119	5.18	S833
8	0.623	0.106	4.02	S833
9	0.687	0.097	3.16	S833
10	0.752	0.089	2.49	S833
11	0.818	0.083	1.93	S833
12	0.883	0.077	1.44	S833
13	0.947	0.072	0.99	S834
14	1.012	0.068	0.59	S834
15	1.077	0.064	0.25	S834
16	1.143	0.060	-0.06	S834
17	1.208	0.057	-0.36	S834
18	1.273	0.054	-0.67	S834
19	1.337	0.051	-0.98	S834
20	1.403	0.048	-1.28	S834
21	1.467	0.047	-1.59	S834
22	1.500	0.046	-1.74	S834

Table 4.11: Blade Geometry of Design Type 1

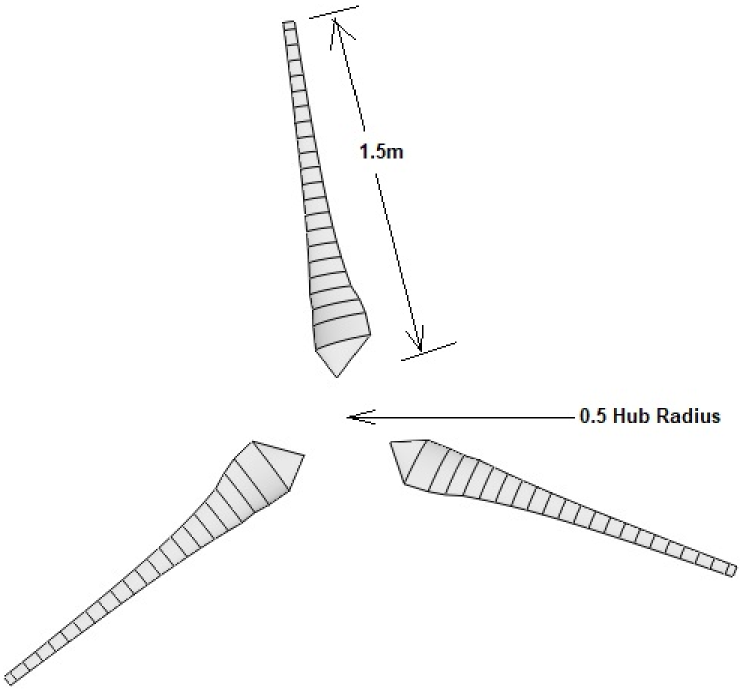

Figure 4.13: Blade design type 1

4.5.2 NREL's Design Type 2 For Rotor Diameter 3-7 meter

According to the section (3.7) NREL suggest to use S823 airfoil at root section of the airfoil, S833 at primary or midsection and S822 at the tip. Using the same blade geometry as shown in section (3.5) and with proposed airfoils, the blade design was developed. Table (4.12) and figure (4.14) illustrates the blade geometry and blade design-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	S823
2	0.233	0.232	21.06	S823
3	0.297	0.199	15.87	S823
4	0.363	0.173	11.92	S823
5	0.427	0.151	8.96	S823
6	0.493	0.133	6.77	S833
7	0.552	0.119	5.18	S833
8	0.623	0.106	4.02	S833
9	0.687	0.097	3.16	S833
10	0.752	0.089	2.49	S833
11	0.818	0.083	1.93	S833
12	0.883	0.077	1.44	S833
13	0.947	0.072	0.99	S834
14	1.012	0.068	0.59	S834
15	1.077	0.064	0.25	S834
16	1.143	0.060	-0.06	S834
17	1.208	0.057	-0.36	S834
18	1.273	0.054	-0.67	S834
19	1.337	0.051	-0.98	S822
20	1.403	0.048	-1.28	S822
21	1.467	0.047	-1.59	S822
22	1.500	0.046	-1.74	S822

Table 4.12: Blade Geometry of Design Type 2

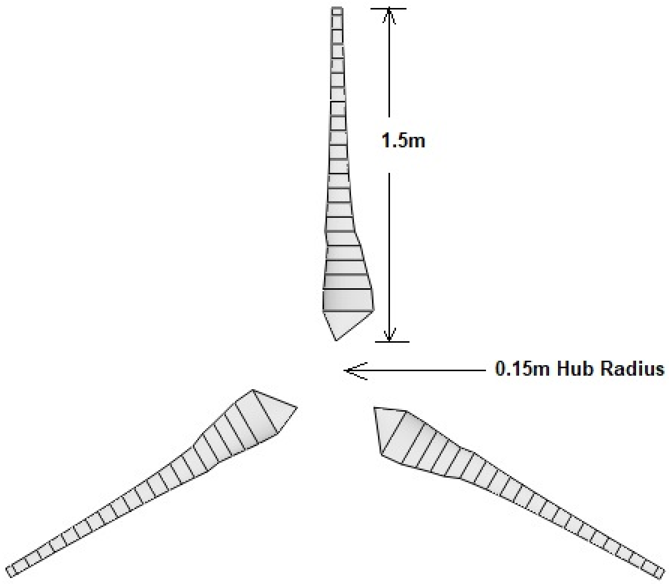

Figure 4.14: Blade design type 2

4.5.3 Performance Analysis of NREL's Design

In this section, we will discuss the performance of the blades. For this, a Cp Vs TSR graph has been generated. We used Anderson et al.(1982) data (geometry and NACA 4412 airfoil) and ran a simulation. With that, a comparison was made with type 1 and type 2 simulation result, to analyze better performing blade. Figure (4.15) illustrates the comparison graph-

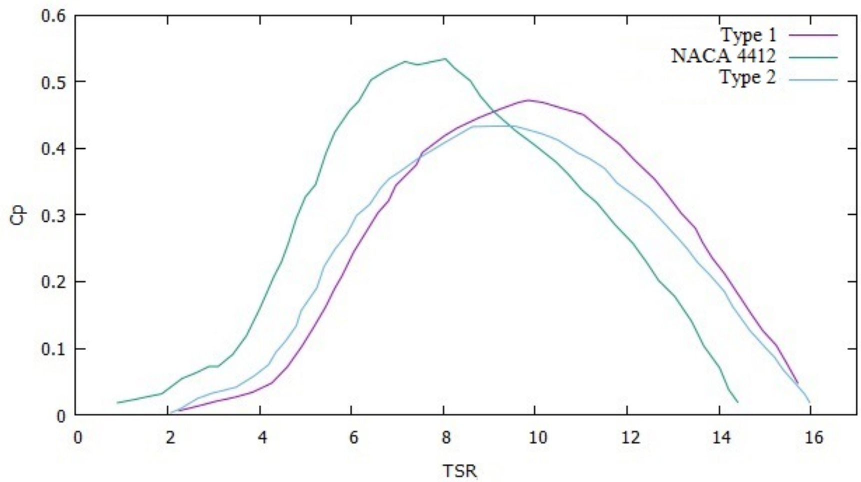

Figure 4.15: Cp Vs TSR Graph comparison

Comparing the graphs of type 1 and type 2 with Anderson et al.(1982) NACA 4412 airfoil graph, we can see that NACA 4412 gives peak Cp value of 0.52 at TSR 8. Cp value is higher for NACA 4412 compared to the other two results till TSR 9. At TSR 9, both NACA 4412 and type 1 design gives the same Cp value of 0.48. Design 2 showing Cp value of 0.43 in that TSR. After TSR 9, design 1 has greater Cp value compared to design 1 and NACA 4412 airfoil. It is notable that, NACA 4412 gives better starting torque compared to the other designs. That is a major advantage for NACA 4412. So overall we can see that NACA 4412 is the better performing blade compared to the other two design.

4.5.4 Blade Design With NACA 2818

The performance result of NREL's two types of blade design was not satisfactory. So we designed blades with NACA airfoils and evaluated blade performance-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	NACA2818
2	0.233	0.232	21.06	NACA2818
3	0.297	0.199	15.87	NACA2818
4	0.363	0.173	11.92	NACA2818
5	0.427	0.151	8.96	NACA2818
6	0.493	0.133	6.77	NACA2818
7	0.552	0.119	5.18	NACA2818
8	0.623	0.106	4.02	NACA2818
9	0.687	0.097	3.16	NACA2818
10	0.752	0.089	2.49	NACA2818
11	0.818	0.083	1.93	NACA2818
12	0.883	0.077	1.44	NACA2818
13	0.947	0.072	0.99	NACA2818
14	1.012	0.068	0.59	NACA2818
15	1.077	0.064	0.25	NACA2818
16	1.143	0.060	-0.06	NACA2818
17	1.208	0.057	-0.36	NACA2818
18	1.273	0.054	-0.67	NACA2818
19	1.337	0.051	-0.98	NACA2818
20	1.403	0.048	-1.28	NACA2818
21	1.467	0.047	-1.59	NACA2818
22	1.500	0.046	-1.74	NACA2818

Table 4.13: Geometry of Bade Using NACA 2818 Airfoil

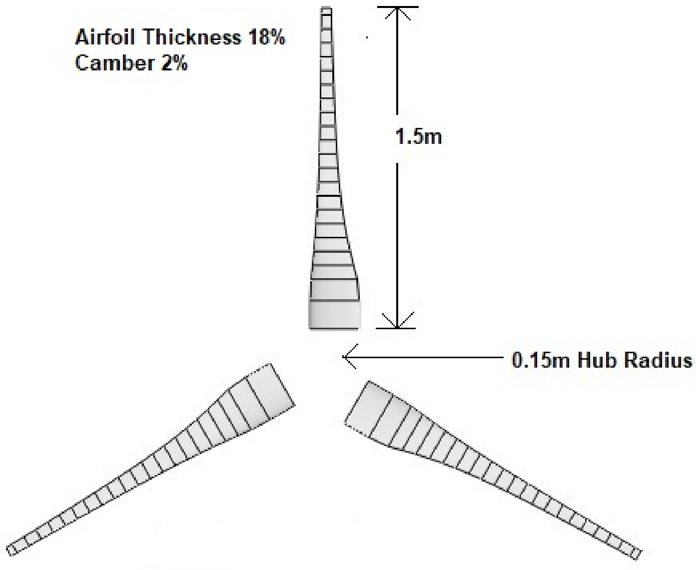

Figure 4.16: Blade design of NACA 2818

4.5.5 Blade Design With NACA 4818

The blade geometry and blade design are shown below-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	NACA4818
2	0.233	0.232	21.06	NACA4818
3	0.297	0.199	15.87	NACA4818
4	0.363	0.173	11.92	NACA4818
5	0.427	0.151	8.96	NACA4818
6	0.493	0.133	6.77	NACA4818
7	0.552	0.119	5.18	NACA4818
8	0.623	0.106	4.02	NACA4818
9	0.687	0.097	3.16	NACA4818
10	0.752	0.089	2.49	NACA4818
11	0.818	0.083	1.93	NACA4818
12	0.883	0.077	1.44	NACA4818
13	0.947	0.072	0.99	NACA4818
14	1.012	0.068	0.59	NACA4818
15	1.077	0.064	0.25	NACA4818
16	1.143	0.060	-0.06	NACA4818
17	1.208	0.057	-0.36	NACA4818
18	1.273	0.054	-0.67	NACA4818
19	1.337	0.051	-0.98	NACA4818
20	1.403	0.048	-1.28	NACA4818
21	1.467	0.047	-1.59	NACA4818
22	1.500	0.046	-1.74	NACA4818

Table 4.14: Geometry of Bade Using NACA 4818 Airfoil

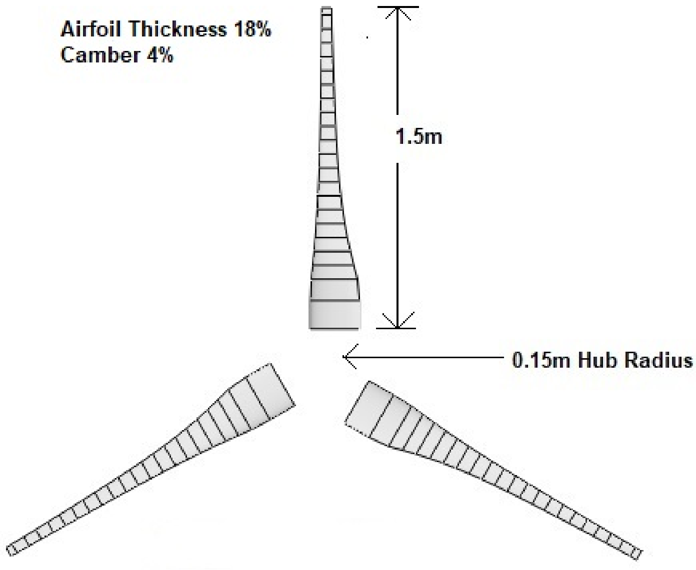

Figure 4.17: Blade design of NACA 4818

4.5.6 Blade Design With NACA 4415

The blade geometry and blade design are shown below-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	NACA4415
2	0.233	0.232	21.06	NACA4415
3	0.297	0.199	15.87	NACA4415
4	0.363	0.173	11.92	NACA4415
5	0.427	0.151	8.96	NACA4415
6	0.493	0.133	6.77	NACA4415
7	0.552	0.119	5.18	NACA4415
8	0.623	0.106	4.02	NACA4415
9	0.687	0.097	3.16	NACA4415
10	0.752	0.089	2.49	NACA4415
11	0.818	0.083	1.93	NACA4415
12	0.883	0.077	1.44	NACA4415
13	0.947	0.072	0.99	NACA4415
14	1.012	0.068	0.59	NACA4415
15	1.077	0.064	0.25	NACA4415
16	1.143	0.060	-0.06	NACA4415
17	1.208	0.057	-0.36	NACA4415
18	1.273	0.054	-0.67	NACA4415
19	1.337	0.051	-0.98	NACA4415
20	1.403	0.048	-1.28	NACA4415
21	1.467	0.047	-1.59	NACA4415
22	1.500	0.046	-1.74	NACA4415

Table 4.15: Geometry of Bade Using NACA 4415 Airfoil

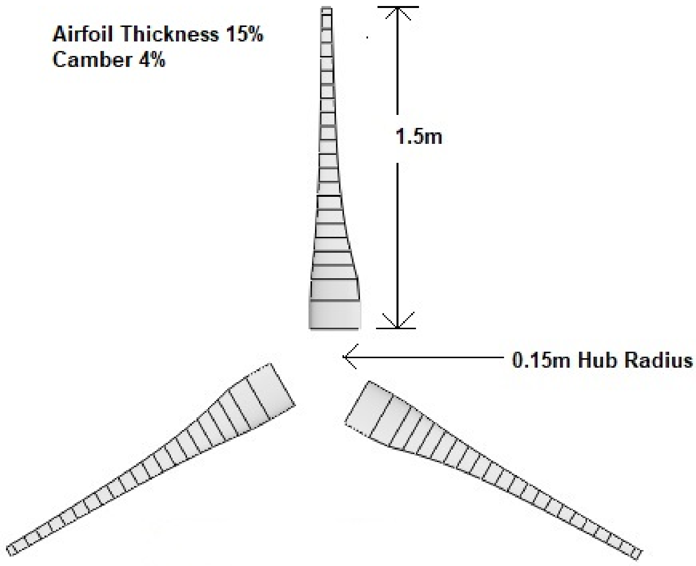

Figure 4.18: Blade design of NACA 4415

4.5.7 Blade Design With NACA 4615

The blade geometry and blade design are shown below-

Element	Position(m)	Chord(m)	Twist(°)	Airfoil
1	0.133	0.250	24.21	NACA4615
2	0.233	0.232	21.06	NACA4615
3	0.297	0.199	15.87	NACA4615
4	0.363	0.173	11.92	NACA4615
5	0.427	0.151	8.96	NACA4615
6	0.493	0.133	6.77	NACA4615
7	0.552	0.119	5.18	NACA4615
8	0.623	0.106	4.02	NACA4615
9	0.687	0.097	3.16	NACA4615
10	0.752	0.089	2.49	NACA4615
11	0.818	0.083	1.93	NACA4615
12	0.883	0.077	1.44	NACA4615
13	0.947	0.072	0.99	NACA4615
14	1.012	0.068	0.59	NACA4615
15	1.077	0.064	0.25	NACA4615
16	1.143	0.060	-0.06	NACA4615
17	1.208	0.057	-0.36	NACA4615
18	1.273	0.054	-0.67	NACA4615
19	1.337	0.051	-0.98	NACA4615
20	1.403	0.048	-1.28	NACA4615
21	1.467	0.047	-1.59	NACA4615
22	1.500	0.046	-1.74	NACA4615

Table 4.16: Geometry of Bade Using NACA 4615 Airfoil

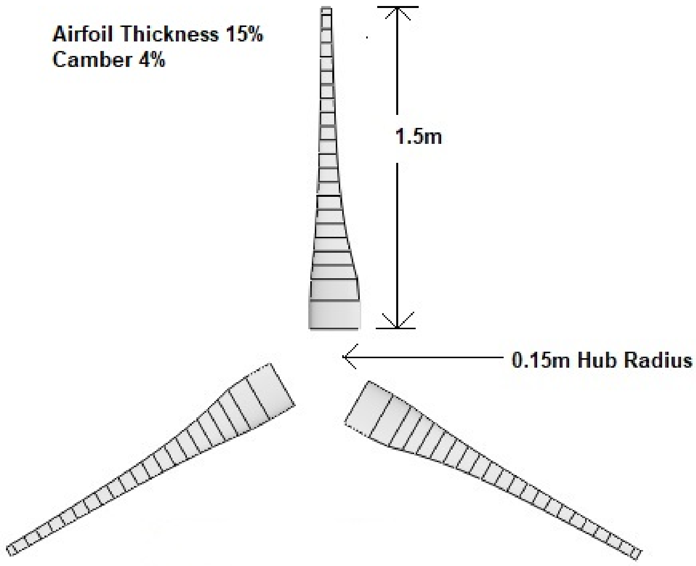

Figure 4.19: Blade design of NACA 4615

4.6 Performance Analysis of NACA Airfoil Design

For performance analysis, the Simulation result of NACA 2818, NACA 4818, NACA 4615 and NACA 4415 have been compared with the Anderson et al. (1982) NACA 4412 simulation result. A Cp Vs TSR graph has been generated. Figure (4.20) illustrates the comparison graph-

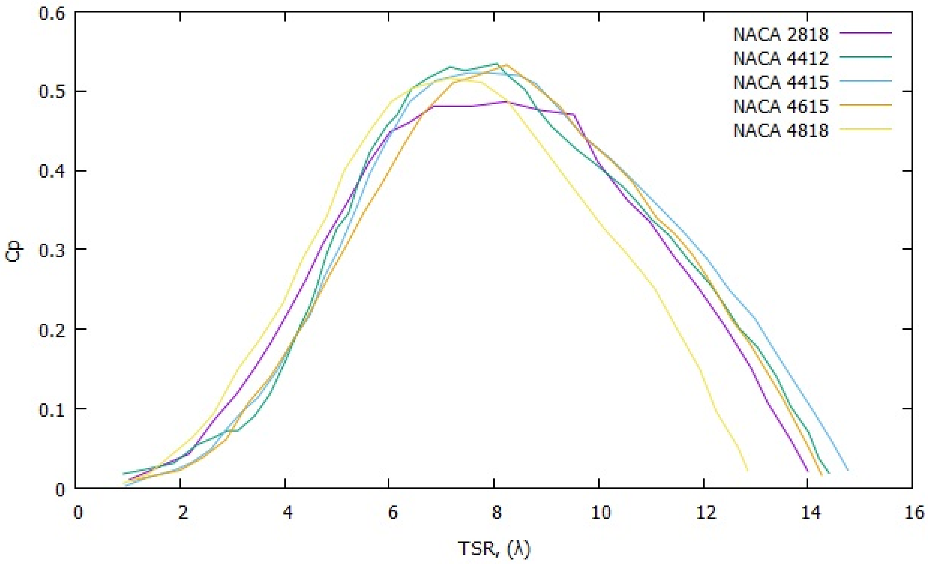

Figure 4.20: Cp Vs TSR Graph comparison

From the figure (4.20) it is an incident that NACA 4818 showing Cp value of 0.47 at TSR 6. Others showing lower Cp value at the same TSR. At TSR range of 6.1-8, this Cp value starts to fall for NACA 4818 and NACA 4412 shows peak Cp value at this range compared to the rest. At 8.1-9.5 TSR range, NACA 4415 and NACA 4615 gives better and similar Cp value compared to other results. From TSR range 11-15 NACA 4415 also shows significant Cp value compared to other results. NACA 2818 doesn't show any significant result for any TSR range. Comparing the starting torque, the NACA 4818 and NACA 2818 shows better result compared to Anderson et al. (1982) NACA 4412 airfoil. So, overall we see NACA 4415, NACA 4615, is best in terms of Cp value. However, the thickness of these airfoils is 15%, whereas NACA 4818 has 18% thickness which is advantageous from the structural point of view. Thicker airfoil helps to withstand the high bending moments causing in structure resulting from lifting force.

4.7 Annual Yield Calculation

The Annual yield of energy production was calculated using the scale and shape parameters calculated in (4.1) for prospective locations with the help of QBlade's "Rotor BEM Simulation" submodule. In the submodule, a turbine can only be defined if a blade is already stored in the database. We have specified all turbine parameters and saved the turbine design. Afterward, a turbine simulation over a range of wind speeds has been conducted. After the input of the k and c values for the Weibull distribution, the annual yield is computed automatically. The required turbine parameters for determining annual yield are shown below-

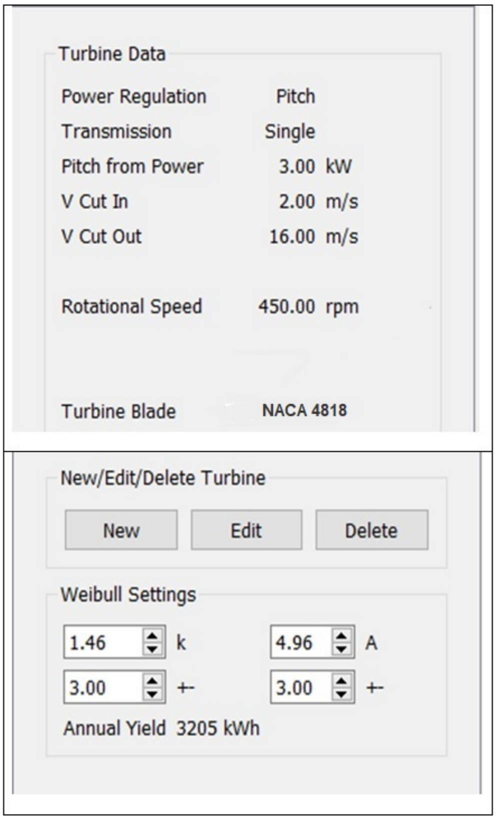

Figure 4.21: Turbine BEM Simulation Module

The annual yields calculated using k and c values are shown below-

Method	Yield Power (kWh)								
	Mongla	Chandpur	Inani	Mymensingh	Parkay	Rangpur	Rajshahi	Shitakunda	Mirzapur
PDM	960	705	2016	269	2107	269	279	818	504
AMLM	2104	1585	2962	1084	3003	1244	1084	1520	1384
SDM	885	618	1852	249	1907	249	246	812	472
MEPM	973	731	631	275	2125	275	325	818	523

Table 4.17: Annual Yield Result of NACA 4818

Method	Yield Power (kWh)								
	Mongla	Chandpur	Inani	Mymensingh	Parkay	Rangpur	Rajshahi	Shitakunda	Mirzapur
PDM	1243	916	2423	396	2519	396	360	1021	683
AMLM	2355	1796	3273	1253	3318	1427	1253	1694	1580
SDM	1165	824	2262	408	2324	373	573	1016	648
MEPM	1256	945	673	404	2536	404	404	1033	695

Table 4.18: Annual Yield Result of NACA 4412

Method	Yield Power (kWh)								
	Mongla	Chandpur	Inani	Mymensingh	Parkay	Rangpur	Rajshahi	Shitakunda	Mirzapur
PDM	1207	912	2450	360	2551	459	350	1014	669
AMLM	2410	1824	3364	1254	3409	1446	1254	1749	1602
SDM	1124	812	2274	336	2221	386	336	1014	610
MEPM	1222	927	615	367	2570	367	370	1032	682

Table 4.19: Annual Yield Result of NACA 2818

Method	Yield Power (kWh)								
	Mongla	Chandpur	Inani	Mymensingh	Parkay	Rangpur	Rajshahi	Shitakunda	Mirzapur
PDM	767	801	2188	308	2282	308	308	899	583
AMLM	2227	1676	3126	1143	3169	1322	1143	1608	1468
SDM	981	709	2020	287	2082	287	487	899	529
MEPM	1072	815	631	315	2300	315	315	916	594

Table 4.20: Annual Yield Result of NACA 4615

Method	Yield Power (kWh)								
	Mongla	Chandpur	Inani	Mymensingh	Parkay	Rangpur	Rajshahi	Shitakunda	Mirzapur
PDM	1168	884	2336	661	2432	361	361	977	655
AMLM	2314	1751	3231	1204	3275	1388	1204	1538	1384
SDM	1090	791	2171	339	2236	339	339	977	599
MEPM	1181	898	673	368	2449	368	368	994	667

Table 4.21: Annual Yield Result of NACA 4415

The calculated result in table 4.17-4.21 shows the annual yield of energy for different locations and methods at 40m height using different NACA airfoil. From the tables, it is seen that Parkay Beach shows the best annual yield result for all the airfoil design followed by Inani. For most of the methods, Inani showing comparatively higher yield exception of MEPM. On the other hand, Parkay also showing comparatively higher yield for all the methods. Locations like Mongla, Chandpur, Shitakunda showing significant results. Mymensingh, Rangpur, Rajshahi, Mirzapur showing a low yield of energy compared to other locations. Overall, Parkay and Inani can be most prospective for harnessing wind energy at 40m height.

Chapter 5

CONCLUDING REMARKS

5.1 Conclusion

In this research, Weibull shape and scale parameters were calculated using four different methods. They are Power density method, alternative likelihood method, standard deviation method, and modified energy pattern factor method. A highly functional programming language, Python was used to solve four different methods numerically for determining the parameters.
A validation of Qlade design with Anderson et al.(1982) experimental work was done and it showed good result.
New turbine blades were designed for performance analysis using NREL's S series airfoils (S822, S823, S833, S834, S835) and NACA airfoils (NACA2818, NACA4818, NACA4415, NACA4615). It is found that Naca 4415 and NACA 4615 are better in terms of Cp value and NACA 4818 is better in terms of its better starting torque and thickness.
Annual yield for prospective locations was calculated. It is found that the Alternative Maximum Likelihood Method gives better annual yield at 40m height for nine prospective locations. Overall Inani, Parky showing better annual yield for most of the methods compared to other locations. So, these two areas can be most prospective for energy production.

5.2 Future Recommendation

The following recommendations are given for the future work-

1. For better shape and scale parameter results, wind speed data needed to be filtered before analysis.
2. For optimum blade design, "blade optimization module" can be used in QBlade.
3. NCrit number should be studied for designing better performing wind turbine blade.

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